Review of AI Video Inpainting Methods with Empirical Benchmarking on a Custom Real-World Dataset

STEPAN GOYUNYAN[1], NAREK MELIKSETYAN[1], ARTHUR TSATURYAN[2], ELIZAVETA LABAZANOVA[3], ZANGIR IKLASSOV[4], (Member, IEEE), and JORGE A. A. HERRERA[4], (Member, IEEE)

[1]Physmath School After Artashes Shahinyan, Yerevan, Armenia (e-mail: stepan.goyunyan@gmail.com; narekmeliqsetyan238@gmail.com)
[2]Evrika STEM Specialized School, Vanadzor, Armenia (e-mail: arturtsaturyan07@gmail.com)
[3]Gymnasium No. 14, Grozny, Chechen Republic, Russia (e-mail: jikitoles@gmail.com)
[4]Mohammad Bin Zayed University of Artificial Intelligence (MBZUAI), UAE (e-mail: zangir.iklassov@mbzuai.ac.ae; jorge.herrera@mbzuai.ac.ae)

Corresponding author: Stepan Goyunyan (e-mail: stepan.goyunyan@gmail.com).

ABSTRACT Video inpainting aims to fill missing or occluded regions in a video so that the result is visually plausible and temporally coherent. This review surveys the evolution from classical patch-based and flow-guided methods to modern generative approaches, including gated convolutional U-Nets, transformer-based models, CLIP-guided editing, and diffusion-based inpainting. We summarize the core ideas, strengths, and limitations of each family and discuss applications in video editing, restoration, film post-production, and autonomous systems. Beyond the survey, we implement and benchmark representative methods, including GAN-based baselines and CLIP-guided variants, on a new real-world dataset containing fast motion, large dynamic objects, and changing illumination. Using standard image-quality, perceptual, and temporal-consistency metrics, we analyze the trade-offs between realism, stability, and compute. Finally, we outline ethical considerations, such as the misuse of seamless content manipulation, and identify open problems and concrete directions toward more robust, efficient, and trustworthy generative video inpainting.

INDEX TERMS Generative artificial intelligence, diffusion models, video inpainting, video object replacement, temporal coherence.

I. INTRODUCTION

VIDEO inpainting is the task of automatically filling missing or damaged regions in a video sequence so that the completed footage looks realistic and plays smoothly over time. This problem appears in many real-world situations: removing unwanted objects from home videos, restoring old damaged films, filling gaps in surveillance footage caused by moving people or vehicles, or completing scenes in movie post-production. What began as a simple image-filling task has become far more challenging in videos because every new frame must connect naturally with the ones before and after it. Over the past fifteen years, video inpainting has evolved rapidly with the rise of artificial intelligence. Early methods relied on copying similar patches from nearby frames or using optical flow to guide the filling process. These traditional approaches worked well for small, static holes but often failed with large moving objects, complex backgrounds, or long sequences. The breakthrough came with deep learning. Starting around 2018, researchers began using convolutional neural networks, generative adversarial networks (GANs), and later transformer models and diffusion models. These generative AI techniques can now understand both the visual content and the motion in a video, producing results that are often indistinguishable from real footage. Generative AI has completely changed what is possible in video inpainting. Modern methods can handle large occlusions, maintain consistent lighting and textures across dozens of frames, and even follow text instructions such as "remove the car and fill the road naturally." Tools built on these ideas are already being used in film editing, autonomous driving (to reconstruct blocked camera views), augmented reality, and video restoration services. At the same time, the same powerful technology raises new concerns: realistic edited videos can spread misinformation, create deepfakes, or violate privacy when applied to surveillance footage. Despite the rapid progress, most existing survey papers focus only on image inpainting or cover methods published before 2023. They rarely provide direct comparisons on the same challenging real-world data. This topical review fills that gap by presenting a complete picture of the field today. We trace the evolution from patch-based methods to the latest generative diffusion and multi-modal approaches. We explain the core ideas behind leading techniques, including gated U-Net networks and CLIP-guided methods. Most importantly, we implemented and compared

TABLE 1. Representative surveys and topic reviews related to image or video inpainting.

Reference	Year	Scope	Taxonomy	Methods emphasized
Ghanbari Talouki *et al.* [1]	2017	Video	Categorizes completion by video content and motion, and by classical families.	PDE-based, texture synthesis, motion-vector and hybrid classical pipelines.
Xiang *et al.* [2]	2023	Image	Multi-perspective taxonomy: strategies, network structures, and loss functions; datasets and metrics.	CNN or GAN-centric deep learning; limited discussion beyond image domain.
Zhang *et al.* [3]	2023	Image	Reviews deep learning image inpainting.	CNN,GAN-heavy.
Quan *et al.* [4]	2024	Image and Video	High-level pipeline taxonomy and architecture families; objectives, datasets, metrics, performance overview.	CNN, VAE or GAN, transformers, diffusion; broad but not implementation-unified on the same real videos.
Elharrouss *et al.* [5]	2025	Image and Video	Transformer-focused taxonomy by architectural configuration, damage type, and metrics; challenges.	Vision Transformers and attention-based designs; diffusion covered but not central.

several state-of-the-art approaches side-by-side on a new custom dataset of real-world videos that we collected ourselves. These videos contain fast motion, large moving occlusions, changing lighting, and complex scenes—conditions that are common in everyday use but missing from many public benchmarks.

A. RELATED SURVEYS AND TOPIC REVIEWS

The rapid evolution of inpainting research has motivated several surveys and topic reviews, but their scope and emphasis vary. Earlier reviews in the *video completion* era largely focused on classical pipelines (e.g., motion compensation, PDE-based propagation) and discussed how assumptions about background or foreground camera motion affect feasibility and visual quality [1]. While these works provide valuable historical context, they predate modern deep generative models that dominate current practice.

More recent surveys primarily target *image* inpainting and propose taxonomies based on (i) inpainting strategy, (ii) network architecture, and (iii) training. For example, Xiang *et al.* organize deep learning inpainting methods across strategy, structure, and loss design, and summarize datasets and evaluation metrics [2]. Zhang *et al.* provide a broad review of deep learning image inpainting, emphasizing architectural families and information fusion patterns, and discuss application-driven task groupings [3]. These image-centric reviews are useful for model design intuition, but they only partially address challenges unique to video, such as long-range temporal consistency, motion drift, and error accumulation. Although [6] presents an AI-powered video editing tool for motion-aware object replacement using generative models, the evaluation is limited in several respects. In particular, the thesis does not provide a systematic comparison with other existing methods, and it does not extensively examine complex scenarios involving heavy occlusion. As a result, it is difficult to fully validate the reported results or assess the proposed method's performance relative to alternative approaches.

Surveys that explicitly cover *both image and video* have appeared very recently and include newer model families. Quan *et al.* provide a comprehensive survey of deep learning-based image and video inpainting, categorizing methods by high-level pipeline and architecture family (CNN,VAE,GAN,transformers,diffusion), and summarizing datasets, objectives, and evaluation protocols [4]. In parallel, Elharrouss *et al.* focus specifically on transformer-based video inpainting, categorizing approaches by architectural configuration, damage types, and performance metrics, and highlighting open challenges and future directions [5]. However, most surveys still synthesize results reported across different papers rather than providing unified, side-by-side implementations on the same difficult real-world videos.

Table 1 summarizes representative surveys and their taxonomies. Fig. 1 provides a compact coverage matrix of which method families are emphasized.

B. PAPER CONTRIBUTIONS

The main contributions of this paper are:

1) A concise overview of video inpainting, from traditional methods to recent generative AI approaches.
2) A comparison of the strengths, limitations, and practical challenges of existing methods.
3) Direct implementation and benchmarking of representative GAN- and CLIP-based methods on a custom real-world video dataset.
4) A discussion of real-world applications and related ethical, privacy, and social concerns.
5) Key open challenges and directions for future research.

II. FOUNDATIONS OF IMAGE INPAINTING

Video inpainting can be viewed as an extension of image inpainting in which, beyond producing a plausible completion for each frame, the method must also enforce *temporal coherence* under motion, occlusions, and long-range dependencies. There are multiple comprehensive surveys covering deep learning architectures, training objectives, benchmark datasets, and evaluation protocols for image inpainting. Readers seeking a broad pipeline- and performance-oriented overview are referred to [7], [8], while [9] emphasizes recent advances in Vision Transformer-based inpainting.

This section therefore does not attempt an exhaustive survey. Instead, it summarizes the image-inpainting concepts that are most frequently reused in video inpainting, such as

Survey	Traditional	CNN/GAN	Transformers	Diffusion	Text / multi-modal
Talouki *et al.* (2017) [1]	Yes	No	No	No	No
Xiang *et al.* (2023) [2]	Brief	Yes	Limited	No	No
Zhang *et al.* (2023) [3]	Brief	Yes	Limited	No	No
Quan *et al.* (2024) [4]	Brief	Yes	Yes	Yes	Limited
Elharrouss *et al.* (2025) [5]	Brief	Yes	Yes	Limited	Limited

FIGURE 1. High-level coverage matrix of representative surveys

priors, *modules*, and *metric choices*, and highlights per-frame failure modes that typically become more pronounced when temporal consistency is required.

A. TASK DEFINITION AND AMBIGUITY

Image inpainting aims to recover visually plausible content in regions where the observation is missing or corrupted [10]. Let $x \in \mathbb{R}^{H\times W\times C}$ denote a corrupted image and $m \in \{0,1\}^{H\times W}$ a binary mask indicating missing pixels. The goal is to predict a completed image $\hat{y}$ that is consistent with the visible context and yields a natural-looking result in the missing region. In supervised learning, a common practice is to construct a corrupted input by replacing masked pixels with a placeholder value and providing the mask to the model:

$$\hat{y} = f_\theta(x, m), \text{ with } x = (1-m)\odot y + m \odot \eta, \tag{1}$$

where y is the ground truth image, η is a placeholder value inserted into masked pixels, and f_θ is a learned model. Variants include blind inpainting [11]–[13], where m is partially unknown or must be inferred, and settings where the corruption is not a clean binary removal (e.g., compression artifacts [14]–[16] or mixed degradations [17], [18]). Regardless of the exact corruption model, inpainting is fundamentally ill-posed: many different completions can match the observed pixels while differing substantially inside the missing region.

This ambiguity, often described as multi-modality of the conditional distribution $p(y|x,m)$ governs the behavior and limitations of inpainting systems. For small holes with strong local cues, plausible completions may cluster tightly (low ambiguity), and reconstruction-style objectives tend to work well [19]–[21]. For large holes or semantically critical regions (faces, text-like structures, unique objects), plausible completions may be diverse (high ambiguity), making a single “correct” answer undefined [22], [23]. Consequently, methods differ in how they trade off fidelity (matching a particular ground truth) versus plausibility (producing any realistic completion), and this trade-off affects both training objectives and evaluation protocols.

This perspective is essential for video inpainting. In the image setting, selecting one plausible completion among many can still yield a satisfactory output. In video, however, the model must make compatible choices across frames: even if each frame is individually plausible, small per-frame variations in structure, texture, or identity can accumulate into temporal artifacts [24]–[26]. For this reason, the remainder of this section treats image inpainting as the conceptual foundation: priors, modules, and objectives; While later sections focus on the additional constraints introduced by temporal coherence.

B. MASK AND CORRUPTION MODELS AS DISTRIBUTIONAL ASSUMPTIONS

Most learning-based inpainting methods are trained under an explicit corruption process that specifies *what* is removed and *how* the model is informed about missingness. In the simplest supervised setting, training pairs are generated by sampling a mask m and forming a corrupted input x from a clean image y, e.g.,

$$m \sim \mathcal{M}, \qquad x = (1-m)\odot y + m\odot\eta, \tag{2}$$

where $\mathcal{M}$ denotes the mask distribution used during training and η is a placeholder value. This perspective highlights an important but sometimes under-emphasized point: *the mask and corruption model are not merely implementation details; they define a distributional assumption*. As a result, generalization is strongly affected by mismatch between the training mask distribution $\mathcal{M}_{\text{train}}$ and the corruption patterns encountered at test time $\mathcal{M}_{\text{test}}$, a theme repeatedly observed across inpainting benchmarks and reviews [7]–[9].

Mask families and inductive bias.

Different mask families emphasize different inductive biases. Regular block masks encourage the model to infer missing content largely from global semantics [19], [27], while irregular masks stress continuity of local structures and boundaries [28]–[31]. Object-like masks require stronger semantic priors and sharper handling of occlusion boundaries. Thin-structure masks are particularly sensitive to geometric priors and high-frequency detail reconstruction. Consequently, reported performance is only meaningful relative to the assumed corruption family; strong results under one mask distribution may not transfer to another.

Mask quality and uncertainty.

In practical pipelines, masks may be produced by upstream segmentation, matting, or tracking modules and can exhibit noisy edges, holes, or temporal jitter. This motivates treating m as a potentially *uncertain* observation rather than a perfect binary indicator. From a modeling standpoint, this includes (i) soft masks, (ii) perturbations, and (iii) blind or partially observed masks where missingness must be inferred jointly with reconstruction. Such deviations often concentrate errors near boundaries, where even minor inconsistencies can cause visually salient seams in images and flicker in videos.

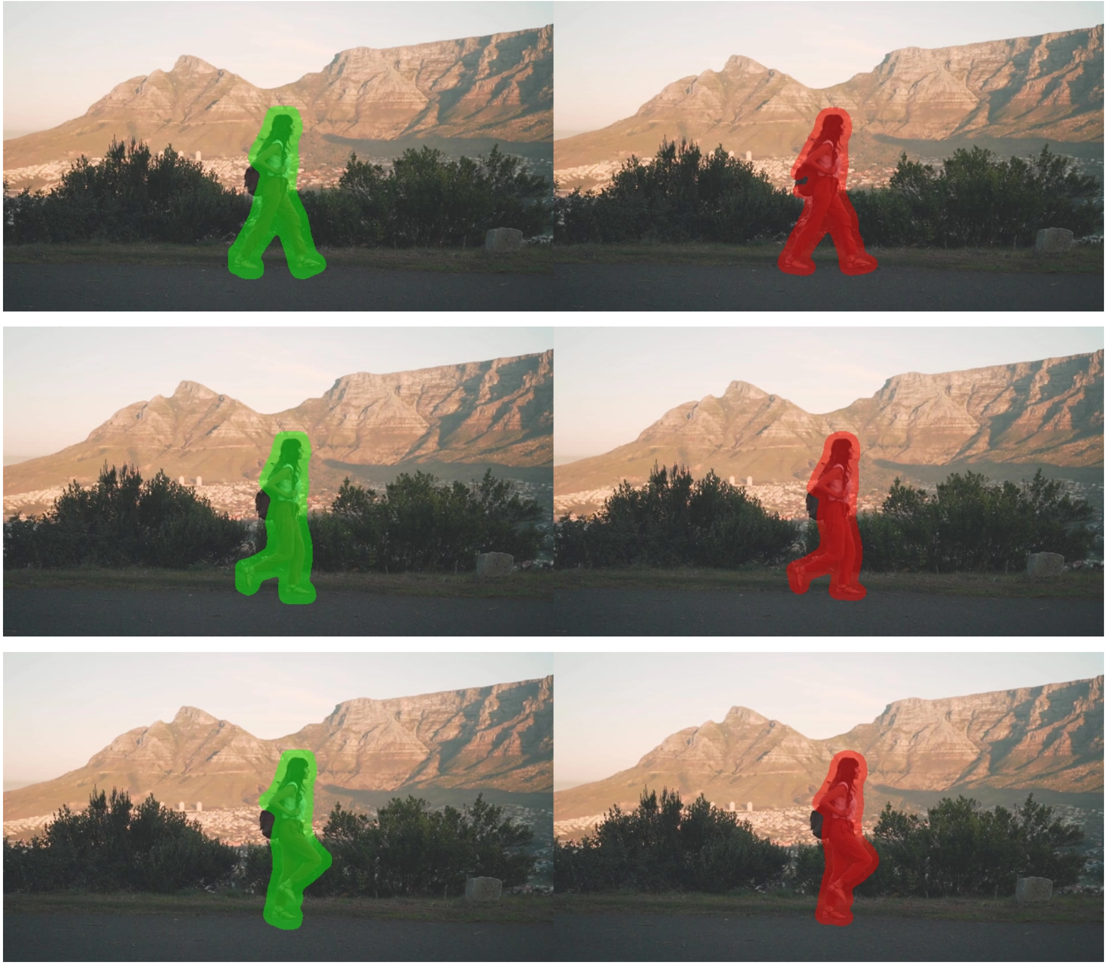

FIGURE 2. **YOLO vs SAM2 Segmentation**

Corruption type	Examples	Parameters ϕ
Binary removal	$x = (1 - m) \odot y + m \odot \eta$	placeholder $p(\eta)$
Partial information	blurred	blur kernel; downsample factor
Compression	JPEG artifacts	quality level
Noise	sensor noise	σ, distribution
Blur	motion	kernel length
Mixed	combinations above	mixture weights

TABLE 2. **Corruption operators, expressed as $x = \mathcal{C}(y; \phi)$.**

Beyond binary removal: broader corruption models.

While binary masking is the dominant formulation, real degradations may combine missing pixels with compression artifacts, motion blur, sensor noise, or overlaid occluders. These cases can be viewed more generally as an operator $x = \mathcal{C}(y; \phi)$ parameterized by corruption variables ϕ (with masks as a special case). We summarize common corruption operators and their parameterizations in Table 2. This approach is useful when occlusion patterns are coupled with motion or when the corrupted observation may contain imperfect, partially informative content rather than pure holes.

Implications for evaluation and for video.

Because the corruption model effectively defines the task, we recommend describing masks and corruptions along a small set of controllable dimensions that can later be instantiated in benchmarks: shape family, coverage ration, connectivity, boundary quality, spatial bias, temporal stability and motion coupling.

In video inpainting, the last factor is particularly critical: even if each frame is inpainted well under a per-frame mask, small mask inconsistencies over time can drive temporal artifacts. Accordingly, later sections revisit these dimensions when discussing correspondence-aware designs (Section III)

Mask dimension	What to report
Shape family	block / free-form / object-like / thin / mixture
Coverage	hole area fraction range
Connectivity	fragmentation level
Boundary quality	binary vs soft; noise; morphology
Spatial bias	placement policy
Temporal stability (video)	mask coherence across frames
Motion coupling (video)	whether mask follows motion

TABLE 3. Recommended reporting for mask distributions $\mathcal{M}$.

and when specifying evaluation protocols once masks and benchmarks are finalized (Section IV).

C. PRIORS: TEXTURE, SEMANTICS, AND STRUCTURE

Although modern inpainting systems are implemented with diverse architectures, many can be understood through the types of *priors* they implicitly or explicitly impose on the solution. These priors determine which completions are preferred among the many possibilities consistent with the visible pixels, and they largely explain typical success cases and failure modes. For the purpose of connecting image and video inpainting, it is useful to group them into three complementary categories: *texture*, *semantics*, and *structure* [7]–[9].

a: Texture prior.

A texture prior assumes that missing regions can be reconstructed by extending or re-synthesizing patterns observed in the known context, leveraging the approximate stationarity of textures and the presence of repeated motifs. Classic patch-based methods operationalize this idea via nearest-neighbor patch search and copying, while deep models learn feature representations that enable non-local matching and texture propagation. Texture priors are effective when the surrounding region contains sufficiently informative exemplars, and they often yield sharp details without requiring strong semantic understanding. However, they are brittle when the missing region contains unique content not present elsewhere in the image, and they may introduce repetitive artifacts or incorrect continuation of patterns across object boundaries.

b: Semantic prior.

A semantic prior favors completions that are consistent with the distribution of natural images and with high-level scene and object statistics. This becomes essential as the missing region grows, where local cues are insufficient and the model must "hallucinate" plausible content. Deep generative inpainting models capture such priors through learned representations that encode object shapes, part configurations, and contextual co-occurrence. The semantic prior improves global plausibility but can also produce semantically inconsistent results when context is ambiguous or when the model over-relies on dataset biases. In such cases, outputs may look realistic in isolation but conflict with the true content or with user intent.

c: Structure prior.

A structure prior emphasizes coherent geometry and sharp boundaries, encouraging the completion to respect contours, perspective, and long-range connectivity of edges and thin structures. In practice, structure can be represented explicitly or implicitly via architectural choices and multi-scale representations that prioritize low-frequency layout before high-frequency detail. Structure priors are particularly important for scenes with strong geometric regularities and for preventing boundary artifacts where the hole meets known pixels. Their limitations are most evident when the structure itself is ambiguous or occluded by large holes, in which case overly rigid structural assumptions can lead to incorrect but confident geometry.

d: Interplay and trade-offs.

These priors are not mutually exclusive: high-quality inpainting typically requires both a plausible global semantic hypothesis and faithful local texture synthesis, constrained by coherent structure. The relative emphasis depends on the mask distribution and context. For small irregular holes, texture and boundary consistency dominate; for large missing regions, semantics and global layout become decisive, often at the cost of increased uncertainty and diversity of plausible outcomes. This also clarifies why different objective functions and evaluation metrics favor different behaviors: pixel-level reconstruction losses tend to average over multiple plausible completions, whereas perceptual or adversarial terms emphasize realism but may tolerate deviations from a particular ground truth.

e: Implications for video inpainting.

In video, these same priors must hold *consistently over time*. Texture priors must avoid temporal "swimming" as patterns are re-synthesized from frame to frame; semantic priors must maintain identity; and structure priors must preserve stable geometry under motion and viewpoint changes. Enforcing such temporally stable priors motivates correspondence-aware mechanisms and spatiotemporal constraints that are central to video inpainting approaches, which we discuss in Section III.

D. REUSABLE DESIGN PATTERNS FROM IMAGE INPAINTING

Despite rapid shifts in backbone architectures, many successful image inpainting methods share a small set of recurring *design patterns*. These patterns are frequently reused inside video inpainting systems as per-frame generators, refinement blocks, or learned priors, with temporal modules added on top. Below we summarize the most common patterns and why they persist across model families [7]–[9].

a: Encoder–decoder backbones with multi-scale context aggregation.

A dominant template is an encoder–decoder generator that compresses the visible context into a global representation and decodes it back to full resolution, typically with skip connections to preserve spatial detail. Multi-scale features are crucial because inpainting requires both low-frequency layout and high-frequency reconstruction. In video pipelines, the same backbone is often applied frame-wise, with temporal information injected via feature warping or spatiotemporal attention (Section III).

b: Mask-aware convolution and feature gating.

Because holes violate the i.i.d. assumptions of standard convolution, many image methods incorporate mask-aware operators to prevent invalid pixels from contaminating features near hole boundaries. Even when specialized convolutions are not used, passing m as additional inputs is a widely adopted pattern. Video methods frequently retain these operators unchanged, since boundary artifacts remain a primary failure mode and are amplified by temporal jitter.

c: Non-local matching for long-range borrowing.

Attention mechanisms allow the model to *copy or reweight* information from distant but relevant regions, approximating the classical ''patch search and copy'' behavior with learned features. This pattern is particularly effective for repeated textures and for propagating consistent appearance across symmetric structures. In video, the same idea extends naturally to cross-frame attention, but the per-frame non-local module remains a useful building block for within-frame self-similarity.

d: Coarse-to-fine generation and explicit refinement.

A recurring strategy is to first predict a coarse, globally plausible completion and then refine details at higher resolution. This can be implemented as a cascade of networks, a multi-resolution decoder, or an explicit ''refiner'' that focuses on boundaries and texture. The motivation is pragmatic: global structure and semantics are easier to stabilize at low resolution, while high-resolution synthesis benefits from a constrained target. Video systems often mirror this organization because temporal errors introduced early can be corrected only if refinement is boundary- and detail-aware.

e: Auxiliary structural guidance.

To strengthen structure priors, many image pipelines predict or consume intermediate representations such as edges, segmentation maps, or depth, and use them to condition the final RGB completion. Even when not explicitly supervised, structure may be encouraged via edge-aware losses or gradient-domain consistency. In video, structural guidance is especially valuable for thin structures and geometry (buildings, text, wires), where temporally inconsistent contours are highly noticeable.

f: Loss decomposition by regions and perceptual realism terms.

Training objectives commonly separate supervision on known or missing pixels, using additional terms on the full image for color consistency. Perceptual losses (feature-space distances) and adversarial losses are frequently added to improve realism and texture sharpness, while total variation or gradient penalties can reduce spurious high-frequency noise near boundaries. These choices carry over to video, but must be revisited: losses that improve single-frame realism can still permit frame-to-frame flicker unless augmented with temporal terms (Section III–IV).

g: Post-processing: blending and boundary handling.

Many pipelines explicitly composite the output with the known region, i.e., $\tilde{y} = (1 - m) \odot x + m \odot \hat{y}$, and may apply additional boundary smoothing or feathering to reduce seams. While simple, such compositing is a reliable safeguard: it guarantees fidelity on observed pixels and isolates model errors to the masked region. In video, the same compositing step remains standard, but boundary instability in m can introduce temporal artifacts, motivating mask stabilization or boundary-aware refinement.

h: Summary: why these patterns persist.

These patterns repeatedly reappear because they correspond to persistent difficulties in inpainting: (i) handling invalid pixels near hole boundaries, (ii) mixing global semantic plausibility with local texture synthesis, and (iii) borrowing information over long spatial ranges. Video inpainting inherits all three and adds the requirement that these mechanisms behave consistently over time; thus, modern video methods typically reuse an image-inpainting core and focus their novelty on how temporal correspondence and long-range propagation are integrated around it.

E. OBJECTIVES AND METRICS AS PROXIES

Because inpainting is intrinsically ill-posed, neither training objectives nor evaluation metrics can be interpreted as direct measures of a unique ''correct'' completion. Instead, they act as *proxies* for desiderata such as (i) fidelity to observed pixels, (ii) plausible synthesis inside the hole, and (iii) (for video) temporal stability. This proxy view is important when comparing methods trained under different mask distributions or when transferring image inpainting components into video pipelines [7]–[9].

a: Region-aware reconstruction losses.
Most supervised objectives start from pixel-wise reconstruction with explicit weighting of missing vs. known pixels:

$$\mathcal{L}_{\text{rec}} = \lambda_{\text{hole}}\big\|m\odot(\hat{y}-y)\big\|_1+\lambda_{\text{known}}\big\|(1-m)\odot(\hat{y}-y)\big\|_1. \quad (3)$$

Reporting hole-only error is common because the known region can dominate full-frame metrics when the missing area is small. Compositing the final output as $\tilde{y} = (1-m)\odot x+m\odot\hat{y}$ additionally guarantees exact agreement with the observed pixels.

b: Perceptual and feature-space objectives.
Pixel losses encourage averaging across multiple plausible completions, often producing overly smooth textures. To better match human judgments of realism, many methods add feature-space losses:

$$\mathcal{L}_{\text{perc}} = \sum_{\ell} w_\ell \left\|\phi_\ell(\hat{y}) - \phi_\ell(y)\right\|_1, \quad (4)$$

where ϕ_ℓ extracts features from a fixed pretrained network. These terms typically improve texture sharpness and global plausibility, but they remain proxies: they may reward "perceptually similar" content that differs from the unknown ground truth, especially for large holes.

c: Adversarial realism terms.
Adversarial losses are commonly used to push completions toward the natural image manifold:

$$\mathcal{L}_{\text{GAN}} = \mathbb{E}\big[\log D(y)\big] + \mathbb{E}\big[\log(1 - D(\hat{y}))\big], \quad (5)$$

often with discriminators focused on the hole region and/or on local patches. GAN terms can improve realism but may introduce hallucinations, mode bias, or training instability. In the proxy interpretation, adversarial terms prioritize "looking real" over matching a specific hidden target.

d: Distributional objectives and stochasticity.
Modern generative formulations further emphasize that the goal is to sample from a conditional distribution rather than regress a single output. In such settings, evaluation becomes more nuanced: metrics penalize diversity, while sample-quality metrics depend on how many samples are drawn and how selection is performed. When image inpainting components are reused in video, this issue interacts with temporal coherence: sampling independently per frame can yield strong per-frame realism but poor stability.

e: Metrics: what they measure.
Common image-level metrics include PSNR and SSIM, LPIPS, and distribution-level scores such as FID. Each metric captures a different axis and can be misleading in isolation:

- **SSIM** favor conservative, averaged predictions and correlate weakly with perceptual realism for large holes.
- **LPIPS** better tracks perceptual similarity but still assumes a single target and may not reflect acceptability under ambiguity.
- **FID-like scores** reflect dataset-level realism but may ignore whether the completion matches the specific scene context.

For inpainting, it is also critical to specify *where* the metric is computed and to report the mask distribution used during evaluation. Table 4 summarizes common inpainting desiderata and the objectives/metrics typically used as proxies, together with characteristic failure modes that often appear when optimizing for (or reporting) each proxy in isolation.

f: Implications for video inpainting.
Video inpainting inherits all image-level proxy issues and adds a new one: a method can score well per-frame while exhibiting temporal artifacts. Consequently, video evaluation typically augments image metrics with temporal proxies. We return to these temporal objectives and metrics in Section IV; here the key takeaway is that *objective/metric choices implicitly define what "good" means*, and those definitions must be aligned with the intended corruption model and the need for temporal coherence.

F. FROM IMAGE ARTIFACTS TO TEMPORAL FAILURE MODES

Many video inpainting failures can be understood as familiar *image* artifacts that become objectionable once they vary over time. While a single-frame completion may appear plausible, even small frame-to-frame inconsistencies are amplified by motion, long-range dependencies, and the human visual system's sensitivity to temporal flicker. This subsection summarizes how per-frame artifacts translate into characteristic temporal failure modes, and links them back to the priors, and proxy metrics (Section II-E) commonly inherited from image inpainting [7]–[9].

a: Boundary artifacts $\rightarrow$ temporal seams.
In images, imperfect blending at the hole boundary manifests as halos, color discontinuities, or slight misalignment of edges. In video, these errors become *temporal seams*: boundary brightness pulsates, edges "breathe", or the boundary appears to shimmer as the mask jitters or the model's boundary prediction changes across frames. This is often driven by (i) temporally inconsistent masks produced by upstream detectors, (ii) region-wise losses that under-weight a narrow boundary band, and (iii) per-frame refinement that does not explicitly enforce boundary stability over time.

b: Texture mismatch $\rightarrow$ swimming, crawling, and popping.
A texture prior that works per-frame (sharp grass, bricks, fabric) may still yield *texture swimming* in video: the synthesized pattern appears to slide over the surface rather than move with it. Related artifacts include *crawling* (high-frequency noise that changes every frame) and *popping* (sudden texture

Goal (what we want)	Metric	Common failure mode)
Preserve observed pixels	Compositing $\tilde{y} = (1-m) \odot x + m \odot \hat{y}$; known-region L_1 or near-zero known-region error	Good known-region fidelity can mask poor hole quality; boundary seams if m is noisy/jittery
Low distortion vs. ground truth	Hole-weighted L_1/L_2; PSNR/SSIM (often hole-only)	Over-smoothing / loss of texture; "average" completions under multi-modality; blurred edges
Perceptual similarity to a reference	LPIPS; perceptual loss $\|\phi(\hat{y}) - \phi(y)\|$	Can reward perceptually similar but incorrect semantics; may ignore fine boundary alignment; can still flicker in video
Photorealism	GAN loss; FID (dataset-level)	Hallucinations inconsistent with context; mode bias; unstable training; realism without correctness
Sharp textures and details	Adversarial + high-frequency/gradient terms; patch discriminators	"Crisp but wrong" textures; repetitive tiling; noisy high-frequency artifacts near boundaries
Structural plausibility	Edge/gradient losses; auxiliary edge/segmentation supervision; boundary-band losses	Structurally plausible but semantically wrong completions; overconfident incorrect geometry on ambiguous holes
Temporal stability (video)	Temporal consistency losses (e.g., flow-warped L_1); temporal LPIPS; video-level FID variants	Over-constraining motion can cause ghosting/trails; flow errors propagate; temporally stable but blurry outputs

TABLE 4. Objectives and metrics act as proxies for different desiderata. Optimizing or reporting a single proxy often yields characteristic failure modes, especially under ambiguity and temporal constraints.

changes when the model switches between plausible modes). These issues are common when textures are generated independently per frame, when patch borrowing is unstable, or when stochastic generative models sample without a mechanism to couple noise across time.

c: Structural errors $\rightarrow$ temporal wobble.

In a single image, a slightly bent line or misconnected contour may be tolerable. In video, these errors produce *wobble*: straight edges (building lines, text baselines, poles, wires) oscillate from frame to frame, and thin structures intermittently break and reconnect. Over longer sequences, the completion may exhibit *geometry drift*, where the shape slowly changes even under nearly static camera motion. Such failures indicate that structure priors are not enforced consistently in time, or that correspondence mechanisms used for propagation are inaccurate near occlusions and thin structures.

d: Semantic ambiguity $\rightarrow$ identity drift and attribute flicker.

Semantic priors help fill large holes, but ambiguity means multiple completions are plausible. In video, if the model resolves ambiguity differently across frames, the result is *identity drift* or *attribute flicker*. This can happen even when per-frame realism metrics are strong, because those proxies do not penalize inconsistent identities unless temporal constraints (or identity-specific losses) are introduced.

e: Propagation and correspondence errors $\rightarrow$ ghosting, trails, and copy artifacts.

Video pipelines often rely on motion (optical flow) or attention-based correspondence to propagate content. When correspondence is wrong—particularly around occlusions, disocclusions, motion blur, and fast motion—errors become temporally structured:

- **Ghosting:** previous content persists as a translucent afterimage due to mis-warping or over-strong temporal smoothing.
- **Dragging:** textures are copied from incorrect regions or frames, producing "stuck" patterns that move inconsistently with the scene.
- **Occlusion boundary errors:** foreground or background layers mix, leading to temporally inconsistent depth ordering.

These artifacts are the temporal counterpart of spatial copy-paste errors in image inpainting, but are often more salient because they track motion.

f: Long-sequence instability $\rightarrow$ error accumulation.

Even if short clips look stable, long videos expose compounding issues: small temporal inconsistencies accumulate into noticeable drift, and occasional failures (bad correspondence, out-of-distribution masks) can trigger *catastrophic switching* where the model abruptly changes its internal hypothesis. This is especially relevant for iterative pipelines and recurrent designs, where errors can feed back into subsequent frames.

g: Why proxy optimization can worsen temporal quality.

Objectives that improve single-frame realism may encourage high-frequency details that are difficult to reproduce consistently over time, increasing flicker. Conversely, strong temporal smoothness penalties can suppress flicker but lead to over-smoothing, ghosting, or lagging updates under rapid motion. This trade-off is a concrete instance of the "proxy" issue discussed in Section II-E: optimizing one proxy can degrade another.

h: Takeaway.

Temporal failure modes are rarely new phenomena; they are typically *image artifacts made dynamic* by motion, mask variation, and correspondence errors. This perspective motivates two practical guidelines for video inpainting design and evaluation: (i) explicitly model and stabilize the sources of temporal variation, and (ii) report metrics and ablations that can separate per-frame quality from temporal consistency. Subsequent sections build on this viewpoint when describing correspondence-aware architectures and temporal objectives, and when defining evaluation protocols for long sequences.

Wexler et al. Space-Time Completion of Video [32]
Granados et al. Background inpainting for videos with dynamic objects and a free-Moving camera [33]
Newson et al. Video Inpainting of Complex Scenes [34]
Hunag et al. Temporally coherent completion of dynamic video [35]

III. VIDEO INPAINTING: INTRODUCING THE TEMPORAL DIMENSION

Deep learning approaches for video object replacement and video inpainting can be organized by (i) how they model *space–time* and (ii) how they enforce *temporal coherence*. In practice, most methods fall into four recurring families:

1) **3D CNN-based methods:** learn spatiotemporal features with 3D convolutions, typically producing stable outputs but sometimes over-smoothing fine details.
2) **Shift-based methods:** propagate or *shift* features through time using learned alignment operators to reuse reliable content from neighboring frames.
3) **Flow-guided methods:** explicitly estimate motion and warp pixels or features to reduce flicker, with known failure modes under occlusions, motion blur, and large displacements.
4) **Attention-based methods:** use global matching to retrieve relevant context across space and time, often improving structural completion at the cost of memory and compute.

This section summarizes these four families and highlights the trade-offs they introduce in practical motion-aware video editing pipelines.

A. GAN AND U-NET-BASED APPROACHES

Core idea. A widely used backbone for image inpainting and editing is a U-Net-style encoder–decoder with skip connections. Let an input frame be $I_t \in \mathbb{R}^{H\times W\times 3}$ and a binary object mask be $M_t \in \{0,1\}^{H\times W}$ (where $M_t(x) = 1$ denotes the region to be synthesized or replaced). A typical masked input is

$$I_t^{\text{mask}} = (1 - M_t) \odot I_t + M_t \odot I^{\text{fill}}, \tag{6}$$

where I^{fill} can be zeros, noise, or a blurred background estimate, and $\odot$ denotes elementwise multiplication.

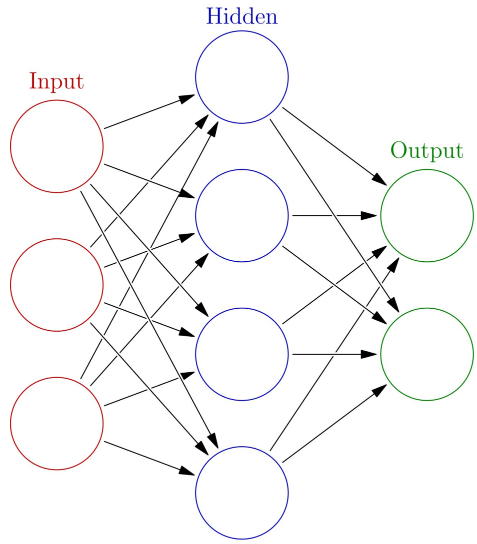

FIGURE 3. Generic schematic of a feed-forward neural network. The thesis pipeline uses multiple learned modules (segmentation/detection, optical flow, diffusion denoising, and conditioning adapters), each implemented as a neural network with task-specific architecture.

Conditional GAN formulation. In conditional GANs for inpainting, the generator G predicts $\hat{I}_t = G(I_t^{\text{mask}}, M_t, c_t)$ given a condition c_t (text, reference image, edges, depth, *etc.*). The discriminator D distinguishes real frames from generated frames (optionally conditioned on the same inputs). A standard objective is

$$\min_G \max_D \ \mathbb{E}\big[\log D(I_t \mid \cdot)\big] + \mathbb{E}\big[\log(1 - D(\hat{I}_t \mid \cdot))\big], \tag{7}$$

which is typically combined with reconstruction and perceptual terms.

Reconstruction and perceptual terms. When a ground-truth target exists, a masked reconstruction loss is

$$\mathcal{L}_{\text{rec}}(t) = \big\|M_t \odot (\hat{I}_t - I_t)\big\|_1. \tag{8}$$

A perceptual loss compares deep features $\phi(\cdot)$:

$$\mathcal{L}_{\text{perc}}(t) = \sum_{\ell} \left\|\phi_\ell(\hat{I}_t) - \phi_\ell(I_t)\right\|_1. \tag{9}$$

Temporal consistency. To reduce flicker, U-Net pipelines often add a warp-consistency constraint using optical flow $F_{t\to t+1}$ and a warping operator $W(\cdot,\cdot)$:

$$\mathcal{L}_{\text{temp}}(t) = \left\|(1 - O_{t+1}) \odot \left(\hat{I}_{t+1} - W(\hat{I}_t, F_{t\to t+1})\right)\right\|_1, \tag{10}$$

where O_{t+1} is an occlusion mask (pixels that cannot be reliably traced from t to $t+1$). In practice, this term stabilizes motion but can penalize legitimate changes. Accurate occlusion handling is therefore critical.

Strengths and limitations. GAN and U-Net models are fast at inference and can be trained end-to-end with explicit temporal regularization. However, they may struggle to synthesize highly detailed and diverse appearances without artifacts, and they can over-smooth textures. For object replacement (as opposed to standard inpainting), the absence of a

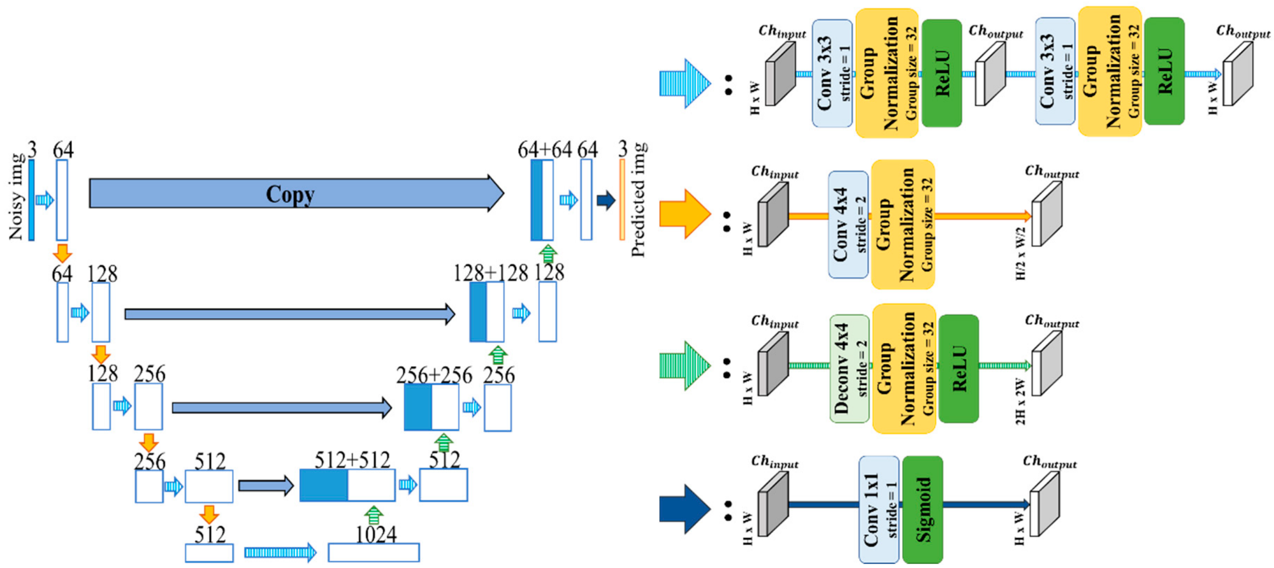

FIGURE 4. Illustrative U-Net style encoder-decoder architecture, commonly used as the denoising backbone inside diffusion pipelines [36]

unique ground truth inside the replaced region also makes purely supervised objectives less straightforward.

B. TRANSFORMER AND ATTENTION MECHANISMS

Motivation. Convolutions are inherently local, whereas attention captures *long-range* dependencies. This is valuable when the missing region requires context from far-away spatial locations or from distant frames.

Self-attention. Given token embeddings $X \in \mathbb{R}^{N\times d}$, attention computes

$$\begin{aligned}\mathrm{Attn}(X) &= \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V,\\ Q = XW_Q, \quad & K = XW_K, \quad V = XW_V.\end{aligned} \tag{11}$$

Spatiotemporal Transformers reshape video features into tokens across (x, y, t), enabling retrieval of relevant patches across frames.

Attention-based inpainting. For video inpainting, attention can be applied (i) within each frame with an explicit temporal module, or (ii) jointly across frames. Joint attention can copy background structure consistently through time, improving completion of repetitive textures and preserving geometric continuity.

Compute and memory. Full attention scales as $\mathcal{O}(N^2)$ in the number of tokens. Since N grows quickly with clip length, practical systems use windowed attention, factorized attention, memory banks, or deformable attention.

Where it fits the four families. Attention-based methods correspond to the *attention-based* family; many strong baselines combine attention with (a) flow guidance or (b) propagation modules, because attention alone can become unstable under fast motion and occlusions.

C. DIFFUSION AND LATENT GENERATIVE MODELS

Forward diffusion. Diffusion models define a noise-adding Markov chain for data x_0:

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\,x_{t-1},\, \beta_t I\right), \quad t = 1, \ldots, T, \tag{12}$$

so that x_T approaches Gaussian noise.

Reverse denoising. A neural network ϵ_θ learns to predict noise given x_t, timestep t, and conditioning c:

$$\min_\theta \; \mathbb{E}_{x_0,\epsilon,t}\left[\|\epsilon - \epsilon_\theta(x_t, t, c)\|_2^2\right], \tag{13}$$

where $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$ and $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$.

Latent diffusion. To reduce cost, latent diffusion encodes an image into a latent $z_0 = E(x_0)$ with a VAE encoder E, runs diffusion in latent space, and decodes with D:

$$z_0 = E(x_0), \qquad \hat{x}_0 = D(\hat{z}_0). \tag{14}$$

Inpainting constraint. For inpainting, one enforces known pixels while sampling unknown regions. A common per-step constraint in image space is:

$$x_t \leftarrow (1-M)\odot x_t^{\text{known}} + M \odot x_t^{\text{gen}}, \tag{15}$$

where x_t^{known} is obtained by noising the original image to the same timestep, and x_t^{gen} is the current sample being denoised. This preserves the background while generating content inside the masked region.

Temporal issues in video. Framewise diffusion often yields high perceptual quality but can flicker because each frame is sampled stochastically. Video-oriented diffusion approaches reduce flicker by (i) reusing latents across frames, (ii) fusing attention across time, or (iii) adding explicit motion constraints and warping-based consistency losses. Diffusion therefore often excels in *appearance realism* while requiring extra mechanisms for *temporal stability*.

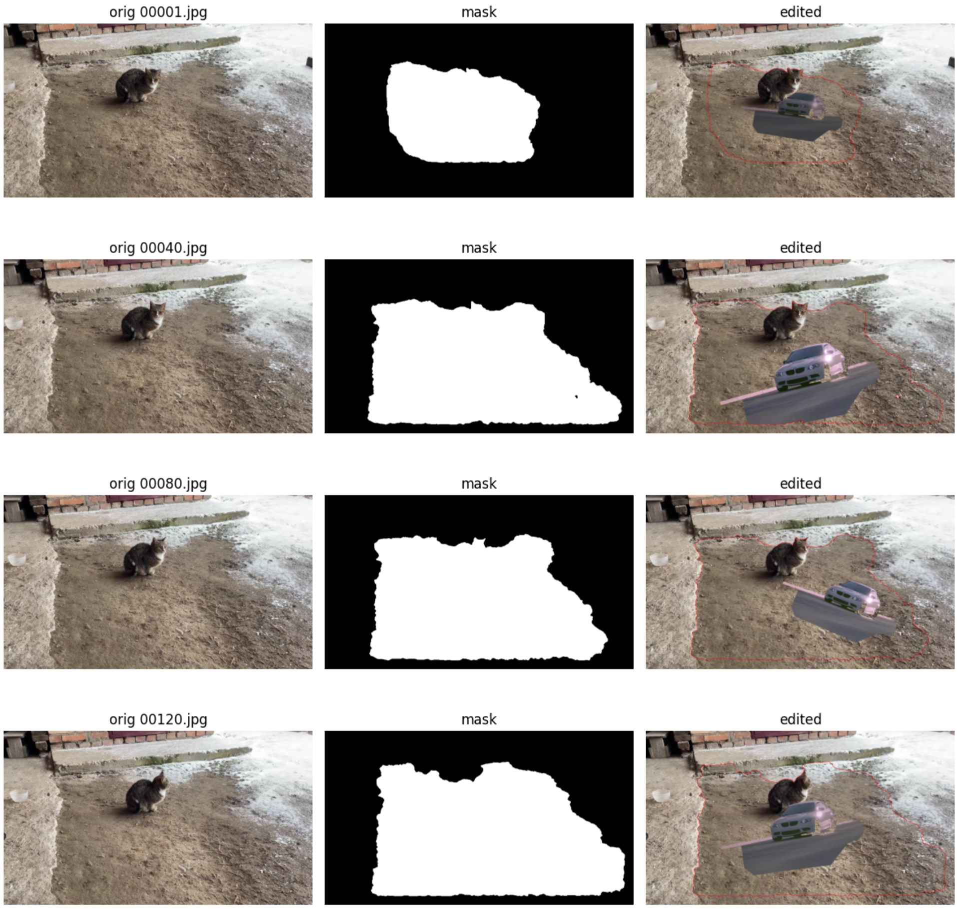

FIGURE 5. **Changing the cat with a car**

D. MULTI-MODAL AND HYBRID METHODS

Why multi-modal? Object replacement is more than ‘‘filling a hole’’: it requires *control* over what is inserted, how it should look, and how it should move with the scene. Multi-modal conditioning provides this control.

General conditioning view. Many hybrid pipelines can be written as a diffusion conditioned on a set of signals:

$$c_t = \left[c_t^{\text{text}},\ c_t^{\text{ref}},\ c_t^{\text{struct}},\ c_t^{\text{motion}}\right], \tag{16}$$

where text provides semantics, reference images provide appearance, structural cues (edges, depth, pose) preserve geometry, and motion cues (flow, trajectories) encourage temporal coherence.

Hybrid composition. A practical modular design is:

1) **Segmentation** to obtain M_t consistently;
2) **Motion module** to propagate an edit prior from t to $t+1$;
3) **Generative module** to synthesize the masked region;
4) **Consistency module** to reduce flicker.

This modular structure is attractive because it reuses strong pretrained components, but it can also accumulate errors (mask drift, flow errors, inconsistent conditioning).

Failure modes. Hybrid systems often fail when (i) masks jitter or lose the target object, (ii) flow is inaccurate under occlusions or blur, (iii) multiple conditions conflict, or (iv) the generative model overfits to the strongest condition and ignores others. These issues motivate careful ablations and explicit evaluation of temporal stability alongside perceptual quality.

IV. IMPLEMENTATION AND COMPARATIVE EVALUATION

V. IMPLEMENTATION AND COMPARATIVE EVALUATION

This section specifies a reproducible and implementable protocol for comparing representative methods under a shared dataset, shared preprocessing, and shared evaluation metrics. The goal is to measure not only per-frame visual quality, but also *temporal consistency*—a primary requirement for video object replacement.

A. CUSTOM VIDEO DATASET

Dataset structure. We define a custom dataset of N short clips:

$$\mathcal{D} = \left\{ \left(\{I_t^{(n)}\}_{t=1}^{T_n},\ \{M_t^{(n)}\}_{t=1}^{T_n},\ a^{(n)} \right) \right\}_{n=1}^{N}, \tag{17}$$

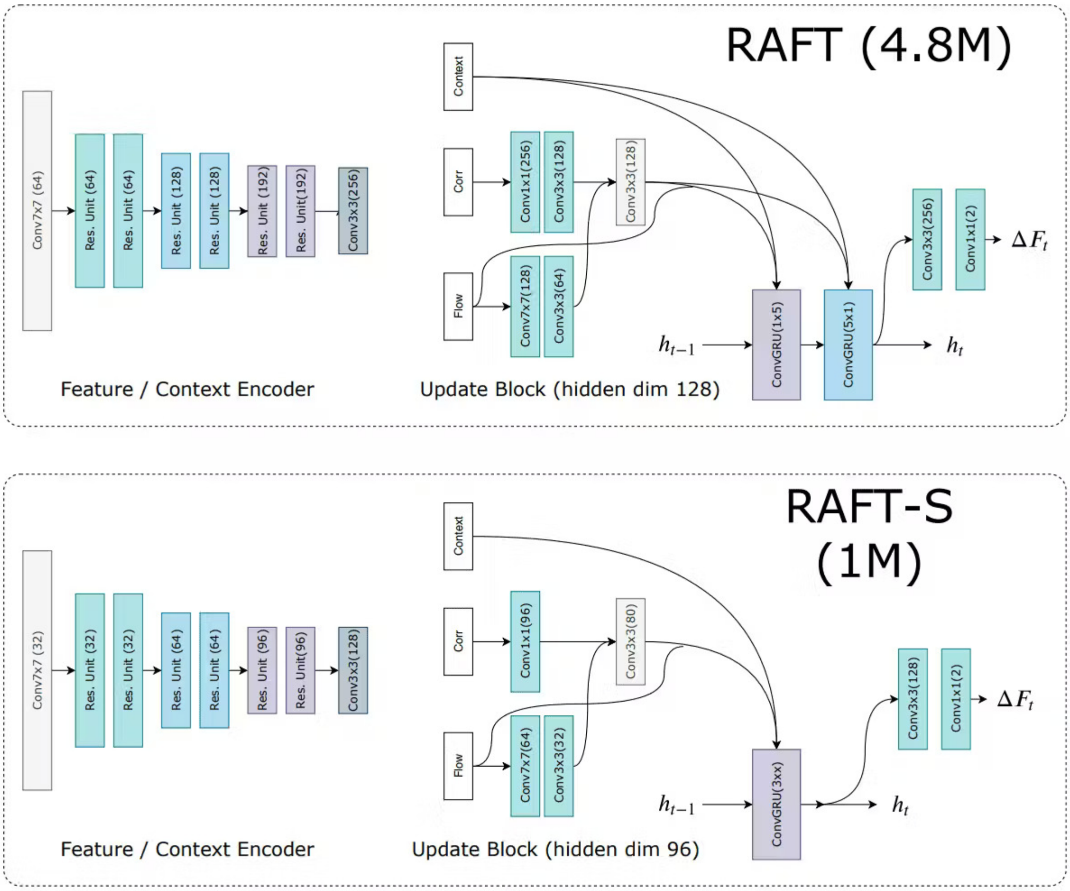

FIGURE 6. High-level schematic of the RAFT optical flow network (and a smaller RAFT-S variant). Conceptually, RAFT extracts features, builds an all-pairs correlation volume, and iteratively refines the flow estimate with a recurrent update operator.

where $I_t^{(n)}$ is the t-th frame, $M_t^{(n)}$ is the object mask, and $a^{(n)}$ stores metadata.

Content diversity. To stress-test temporal coherence, the dataset should include:

- **Rigid objects** and **non-rigid objects**.
- **Camera motion** and **static-camera** clips.
- **Occlusions**.
- **Lighting changes**.

Mask generation. Masks $\{M_t\}$ can be produced by (i) per-frame segmentation, (ii) semi-automatic keyframe annotation with propagation, or (iii) a tracking-based video segmentation model. Because mask jitter directly causes flicker, we recommend reporting a simple mask-stability statistic:

$$\Delta_M(t) = 1 - \frac{|M_t \cap M_{t-1}|}{|M_t \cup M_{t-1}|}, \tag{18}$$

which is the frame-to-frame Jaccard distance.

Splits and preprocessing. We split $\mathcal{D}$ into train, validation and test by clip to avoid leakage. Frames are resized to a fixed resolution, normalized to $[0, 1]$, and sampled at a consistent frame rate. For methods requiring fixed-length windows, we sample sub-clips of length L with overlap.

B. SELECTED METHODS AND IMPLEMENTATION DETAILS

Coverage across method families. To reflect the four directions and modern generative editing, we select a small but representative set:

- **Trajectory baseline:** similarity-transform propagation of a replacement prior across frames.
- **Flow-guided baseline:** optical-flow warping + inpainting of occluded regions.
- **Attention baseline:** spatiotemporal attention-based inpainting over a temporal window.
- **Diffusion baseline:** per-frame latent diffusion inpainting with optional temporal controls.

The comparison is designed to isolate trade-offs: realism vs. stability vs. compute.

Motion propagation. Let $F_{t\to t+1}$ be forward flow and $W(\cdot, F)$ be the forward warp. A propagated prediction from frame t into $t+1$ is

$$\tilde{I}_{t+1} = W(\hat{I}_t, F_{t\to t+1}). \tag{19}$$

To avoid copying across occlusions, define an occlusion mask O_{t+1}. One practical choice is forward–backward consistency. Let $F_{t+1\to t}$ be backward flow; a pixel is unreliable if

$$\left\| F_{t\to t+1}(x) + F_{t+1\to t}\big(x + F_{t\to t+1}(x)\big) \right\|_2 > \tau. \tag{20}$$

FIGURE 7. Car transition images

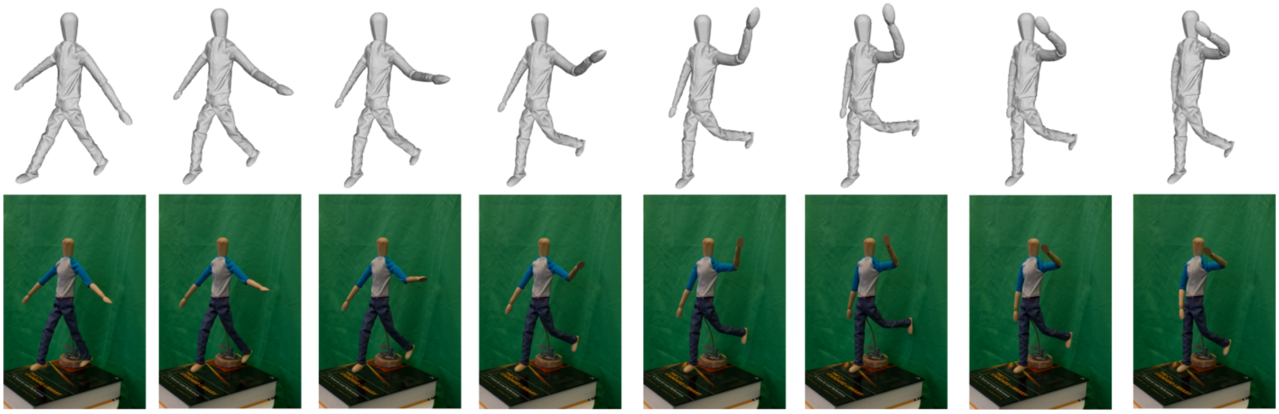

FIGURE 8. Example of articulated human motion across frames. Such motion changes internal configuration in ways a single global similarity transform cannot represent

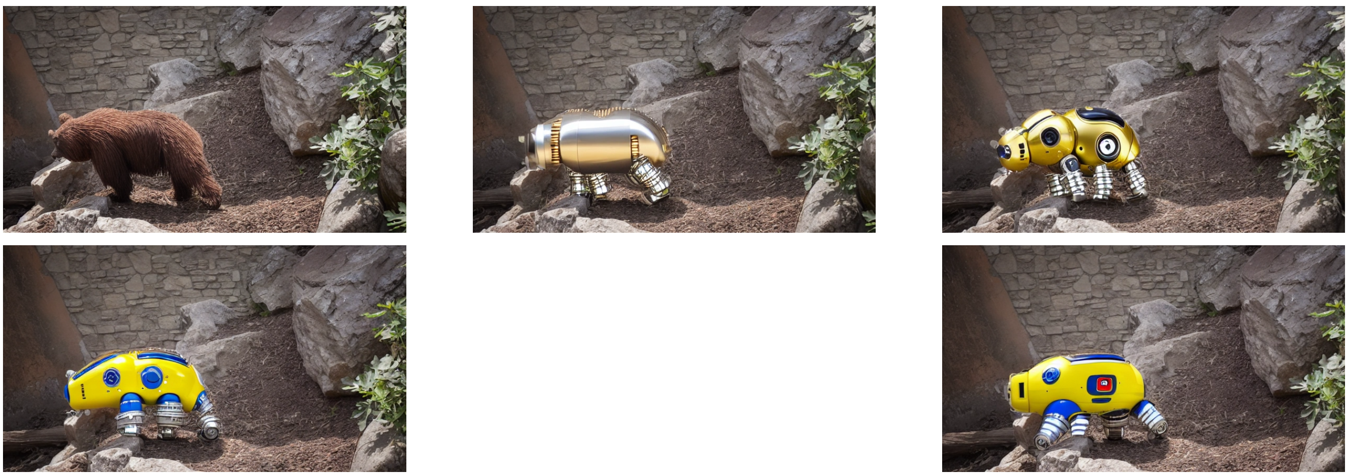

FIGURE 9. Transition process of a bear image

Then we blend the propagated prediction with the original frame in unreliable regions:

$$\bar{I}_{t+1} = (1 - O_{t+1}) \odot \tilde{I}_{t+1} + O_{t+1} \odot I_{t+1}. \tag{21}$$

Finally, a generative inpainting model fills the target region using mask M_{t+1}:

$$\hat{I}_{t+1} = \mathcal{G}(\bar{I}_{t+1}, M_{t+1}, c_{t+1}), \tag{22}$$

where $\mathcal{G}$ can be a U-Net, Transformer, Diffusion inpainting module. **Diffusion with temporal initialization.** For diffusion-based editing, a simple temporal trick is to initialize the sample for frame $t+1$ from a warped latent of frame t:

$$x_T^{(t+1)} \leftarrow \eta\,\mathcal{N}(0, I) + (1 - \eta)\,W(\hat{I}_t, F_{t \to t+1}), \tag{23}$$

with $\eta \in [0, 1]$ controlling how strongly we trust the warped prior. This can reduce flicker, but it may introduce stretching artifacts if the flow is inaccurate.

Implementation reporting. For reproducibility, each method should report: input resolution, frame rate, clip window length L, number of model parameters, GPU type, peak memory, and average runtime. If prompts or reference images are used, list them explicitly and keep them fixed across methods.

C. EXPERIMENTAL RESULTS

Frame and video aggregation. For any per-frame metric m_t, we report per-video mean and standard deviation:

$$\mu_m = \frac{1}{T}\sum_{t=1}^{T} m_t, \qquad \sigma_m = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(m_t - \mu_m)^2}. \quad (24)$$

Dataset-level numbers are computed by averaging μ_m across videos.

Pixel-level fidelity (background preservation). For object replacement, ground truth inside the replaced region may not exist. A practical alternative is to measure distortion on the *unchanged background* region $U_t = \Omega \setminus M_t$:

$$\mathrm{MSE_{bg}}(t) = \frac{1}{|U_t|}\sum_{x\in U_t}\left\|\hat{I}_t(x) - I_t(x)\right\|_2^2, \qquad (25)$$

$$\mathrm{PSNR_{bg}}(t) = 10\log_{10}\left(\frac{\mathrm{MAX}^2}{\mathrm{MSE_{bg}}(t)}\right), \qquad (26)$$

where $\mathrm{MAX} = 1$ for $[0,1]$ images. We also compute full-frame and edit-region PSNR/SSIM/MAE, where the edit region is the masked object area.

Temporal consistency. Using backward flow $F_{t+1\to t}$, define a warping error over non-occluded pixels $\Omega_{t+1}^{\mathrm{vis}}$:

$$E_{\mathrm{warp}}(t) = \frac{1}{|\Omega_{t+1}^{\mathrm{vis}}|}\sum_{x\in\Omega_{t+1}^{\mathrm{vis}}}\left\|\hat{I}_{t+1}(x) - \hat{I}_t\big(x + F_{t+1\to t}(x)\big)\right\|_1. \qquad (27)$$

Lower temporal error indicates less flicker under motion.

Efficiency metrics. Practical deployment metrics include:

$$\mathrm{FPS} = \frac{\#\text{frames}}{\text{total inference seconds}}, \qquad (28)$$

plus parameter count, FLOPs/GFLOPs, and peak GPU memory.

Evaluation setting. We evaluate three implemented variants: (i) a GAN-based replacement baseline, (ii) a CLIP-guided variant, and (iii) a Stable-Diffusion-based video replacement pipeline with SAM2 masks. The GAN and CLIP variants are evaluated on 18 clips covering cats, cars, and humans. The Stable Diffusion setting is evaluated on seven 60-frame trials, consisting of three car trials, three cat trials, and one human trial. Unless stated otherwise, the reported values are obtained by first averaging each metric over frames in a video and then averaging over videos.

Metric interpretation. Full-frame metrics quantify global similarity to the input video. Background metrics are computed outside the edited mask and therefore measure how well the unchanged scene is preserved. Edit-region metrics are computed inside the object mask and should be interpreted differently: because object replacement intentionally changes the masked content, lower edit-region similarity does not necessarily mean a worse edit; instead, it indicates a larger deviation from the original object. Conversely, extremely high edit-region similarity can indicate that the original object was preserved rather than successfully substituted. This distinction is especially important for the CLIP-guided variant, which produced very low errors because it did not consistently replace the target objects in the videos. Temporal metrics measure frame-to-frame consistency, while runtime metrics indicate practical deployment cost. For the CLIP-guided variant, several PSNR values are infinite because some regions are reconstructed exactly; therefore, PSNR averages for that method are computed over finite values only.

Table 5 shows that the three methods behave differently under the proposed benchmark. The GAN baseline obtains moderate full-frame reconstruction quality, with an average full-frame PSNR of 23.56 dB and SSIM of 0.645. Its background PSNR is higher than its full-frame PSNR, indicating that much of the error is concentrated around the edited object and mask boundary. The edit-region scores are also modest, which is expected for an object-replacement task but still indicates that the synthesized region differs substantially from the original masked content.

The CLIP-guided variant achieves near-perfect numerical similarity across most metrics. Its full-frame SSIM is 0.99998, the average LPIPS is close to zero, and both background and edit-region MAE are extremely small. However, qualitative inspection shows that this is not evidence of successful object substitution. Instead, the CLIP-guided variant often keeps the original object or changes it only weakly, so the output remains very close to the input. The low error therefore reflects conservative frame preservation rather than effective semantic replacement.

The Stable Diffusion + SAM2 pipeline produces the opposite behavior and gives the best visual object-replacement results in our experiments. It preserves the background reasonably well, with an average background PSNR of 28.11 dB, but its edit-region metrics are much lower than the other two methods. This is expected because the diffusion model actually replaces the target object inside the masked region, so the generated content is intentionally far from the original object. Thus, the lower edit-region PSNR/SSIM should not be interpreted as worse editing by itself; rather, it confirms that the method performs a stronger semantic edit. At the same time, the high edit-region MAE and low edit-region SSIM show that maintaining local structure, identity, and boundary consistency remains challenging.

Table 6 summarizes the temporal and efficiency behavior. The CLIP-guided variant has the smallest measured temporal error, again supporting the observation that it remains very close to the input frames rather than reliably substituting the object. The GAN baseline has substantially larger temporal error, which is consistent with flicker and boundary drift in challenging clips. The Stable Diffusion + SAM2 pipeline achieves a mean temporal L1 of 0.0336 and temporal SSIM of 0.766 across the seven trials. Although these temporal scores are not the lowest, the qualitative outputs show the most

TABLE 5. Main quantitative comparison across the evaluated methods. PSNR is reported in dB, MAE is reported on the 0–255 intensity scale, and LPIPS is lower better. For CLIP, PSNR values are finite-value means because some samples produced infinite PSNR. The CLIP scores should be read as preservation scores rather than substitution success, because qualitative inspection showed that CLIP often left the original object largely unchanged.

Method	Videos	Frames/video	Full PSNR ↑	Full SSIM ↑	BG PSNR ↑	BG MAE ↓	Edit PSNR ↑	Edit SSIM ↑	Edit MAE ↓	LPIPS ↓
GAN baseline	18	200	23.56	0.645	26.28	9.42	16.84	0.472	27.86	0.4949
CLIP-guided variant	18	120	84.59	0.99998	82.49	0.00032	82.26	0.99997	0.00076	0.000006
Stable Diffusion + SAM2	7	60	20.61	0.698	28.11	8.41	9.85	0.268	62.50	–

TABLE 6. Temporal consistency and efficiency. The temporal error for GAN and CLIP is the background flow MAE used in the framewise evaluation, whereas Stable Diffusion reports frame-to-frame temporal L1 and temporal SSIM from the video-level JSON summaries; these temporal numbers are therefore complementary rather than directly identical units.

Method	Temporal error ↓	Temporal SSIM ↑	Runtime / evaluation time	Seconds per frame	Observation
GAN baseline	5.878	–	113.06 s per evaluated clip	–	Faster evaluation, but larger temporal error and visible instability in difficult categories.
CLIP-guided variant	0.00028	–	84.33 s per evaluated clip	–	Very low error because the original object is often preserved.
Stable Diffusion + SAM2	0.0336	0.766	6597.46 s per 60-frame trial	109.96	Best visual substitution, but computationally expensive.

TABLE 7. Category-level Stable Diffusion + SAM2 results. The values show that difficulty varies substantially by object type and scene dynamics.

Category	Trials	Full SSIM ↑	BG PSNR ↑	Edit SSIM ↑	Temp. L1 ↓	Temp. SSIM ↑
Cars	3	0.746	28.84	0.258	0.0475	0.653
Cats	3	0.605	25.48	0.228	0.0258	0.850
Humans	1	0.837	33.83	0.417	0.0153	0.851

convincing object replacement among the evaluated methods. Its runtime is high: approximately 6597 seconds per 60-frame trial, or about 110 seconds per frame. This makes the current diffusion pipeline useful as a high-quality research baseline but not yet suitable for real-time or interactive editing.

The category-level diffusion results in Table 7 show that cars are the most temporally difficult category in the Stable Diffusion setting, with the highest temporal L1 and the lowest temporal SSIM. Cat replacement is more temporally stable but has lower full-frame and edit-region SSIM, suggesting appearance drift and structural mismatch inside the generated object. The human-to-drone case obtains the strongest background preservation and edit-region SSIM among the diffusion trials, although it is represented by only one trial and should therefore be interpreted cautiously.

D. ANALYSIS AND INSIGHTS

Background preservation is easier than object replacement. Across the evaluated methods, background metrics are generally stronger than edit-region metrics Fig. 11. This is expected because the background is mostly copied or preserved from the original video, while the masked region requires synthesis. The Stable Diffusion + SAM2 pipeline illustrates this most clearly: it reaches 28.11 dB background PSNR but only 9.85 dB edit-region PSNR. This gap confirms that the main challenge is not simply preserving the unedited scene, but producing a replacement that remains structurally plausible, semantically correct, and temporally stable inside the mask.

High numerical similarity does not imply successful object substitution. The CLIP-guided variant obtains the best PSNR, SSIM, LPIPS, and temporal-error values, but qualitative inspection shows that it does not reliably substitute the target objects. These values are therefore best interpreted as evidence of frame preservation, not as evidence of successful replacement. In object replacement, a method can score well by changing very little or by leaving the original object in place. For this reason, pixel similarity metrics must be paired with qualitative examples and, in future work, target-fidelity metrics that explicitly measure whether the requested replacement object appears with the correct identity, pose, and appearance.

Stable Diffusion gives the best visual replacement but exposes the limits of current metrics. Stable Diffusion produces the most convincing visual substitutions among the evaluated methods because it actually generates new object content inside the SAM2 mask. Its lower edit-region PSNR and SSIM therefore partly reflect successful semantic change rather than failure. At the same time, the method is computationally expensive and still exhibits temporal instability in some sequences Fig. 12. The results suggest that diffusion-based replacement is the most promising direction for realism and controllability, but it requires stronger temporal conditioning. Without robust temporal control, the model may generate plausible frames independently while allowing the object's appearance, texture, or identity to drift over time.

Object category affects the bottleneck. The category-level results show that different objects fail in different ways. Cars produce the weakest temporal scores in the diffusion trials, likely because rigid object replacement is sensitive to perspective, scale, and motion alignment. Cats show better temporal scores but lower structural similarity, consistent with the difficulty of replacing deformable animals while preserving pose and fur structure. The human-to-drone trial performs best numerically, but because it contains only one trial, it should be treated as an illustrative case rather than a category-level conclusion.

Typical trade-offs. The comparison supports a realism–stability–compute trade-off, but it also shows that low error is not sufficient for object replacement. The GAN baseline is computationally more practical but produces weaker edit-region quality and temporal stability. The CLIP-guided variant is numerically stable because it often fails to make a substantial substitution. The Stable Diffusion + SAM2 pipeline provides the best visual replacement and enables stronger semantic editing, but it is expensive and still needs improved

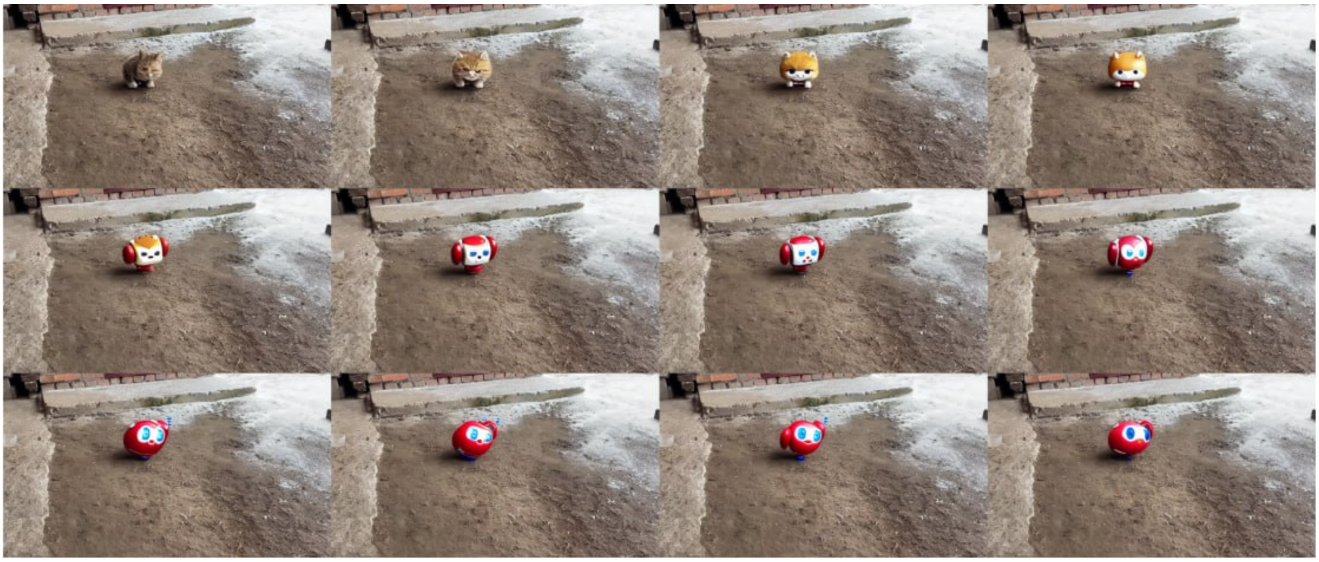

FIGURE 10. Supplementary qualitative result from a reviewer-run notebook execution. Across the 12-frame sequence, the replacement remains near the intended scene region, but its appearance drifts strongly over time, changing color, shape, and apparent identity. The grid therefore supports the review's central claim that coarse motion transfer or location consistency alone does not guarantee temporal appearance consistency.

temporal identity locking. In many comparisons, methods align along the following axes:

- **3D CNN** approaches often improve stability but may blur fine textures.
- **Flow-guided** methods can be sharp and stable when flow is accurate, but degrade under occlusions, fast motion, or motion blur.
- **Attention-Transformers** better preserve global structure and repeated textures, but can be memory-intensive and may still flicker if not explicitly constrained.
- **Diffusion** often yields strong per-frame realism, but needs temporal controls to avoid framewise randomness and identity drift.

A unified view via objectives. Many training-based methods can be interpreted as minimizing a weighted sum:

$$\mathcal{L} = \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{perc}}\mathcal{L}_{\text{perc}} + \lambda_{\text{temp}}\mathcal{L}_{\text{temp}} + \lambda_{\text{adv}}\mathcal{L}_{\text{GAN}}, \tag{29}$$

where $\mathcal{L}_{\text{temp}}$ is usually the deciding factor for flicker reduction. The present results suggest that future systems should also include target-fidelity or identity-locking terms, especially for controllable object replacement where the desired output is not merely any plausible inpainting but a specific replacement object.

Failure analysis checklist. For each method, failures can be attributed to one of four sources:

1) **Mask instability** $\rightarrow$ drifting boundaries and popping artifacts.
2) **Motion estimation errors** $\rightarrow$ stretching, ghosting, and incorrect perspective.
3) **Generative variability** $\rightarrow$ identity changes across frames.
4) **Condition conflicts** $\rightarrow$ wrong geometry, wrong appearance, or incomplete replacement.

Reporting metrics together with these diagnoses makes the comparison actionable: it clarifies whether the bottleneck is segmentation, motion estimation, conditioning, or the generator itself. Overall, the experiments indicate that robust video object replacement requires all four components: stable masks, reliable motion alignment, strong generative synthesis, and explicit temporal identity control.

VI. SOCIAL, ETHICAL, AND PRIVACY IMPLICATIONS

As generative AI video inpainting tools become easier to use, their power to create realistic footage raises important questions about how society will be affected. The same technology that can clean up old family videos or remove a distracting object from a vacation clip can also be misused to alter reality in ways that harm individuals, organizations, and public trust.

A. THE PERVASIVE ROLE OF GENERATIVE VIDEO INPAINTING

Video inpainting is now part of everyday tools used by filmmakers, content creators, and even smartphone apps. What started as a technical solution for damaged footage has become a creative and practical feature in video editing software. Millions of users can now remove unwanted people, add missing objects, or change backgrounds with just a few clicks. This widespread availability means the technology touches many areas of life, from social media to news reporting and legal evidence.

B. RISK OF MISINFORMATION AND DEEPFAKES

One of the biggest concerns is the creation of convincing fake videos. Generative models such as diffusion-based inpainting or CLIP-guided methods can fill large missing areas so naturally that viewers cannot tell the difference from real footage. A short clip of a public figure saying or doing something

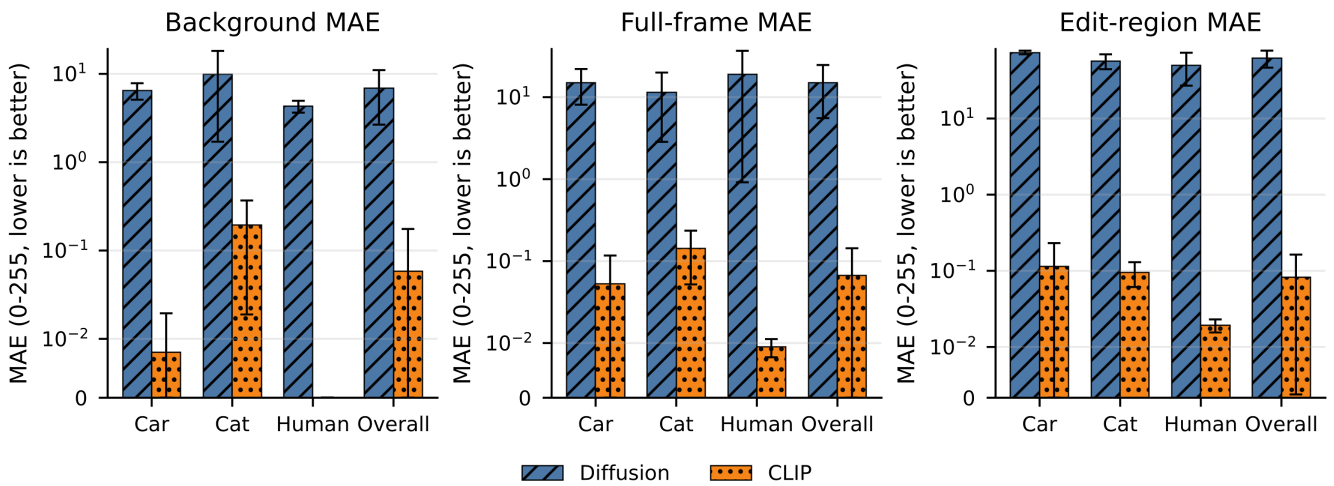

FIGURE 11. Category-wise MAE comparison for the matched seven-sequence experiment. Bars report the video-level macro mean and error bars report the sample standard deviation. All MAE values use the 0–255 intensity scale, and lower values are better. The near-zero CLIP errors primarily indicate preservation of the source appearance and should not, by themselves, be interpreted as successful object substitution.

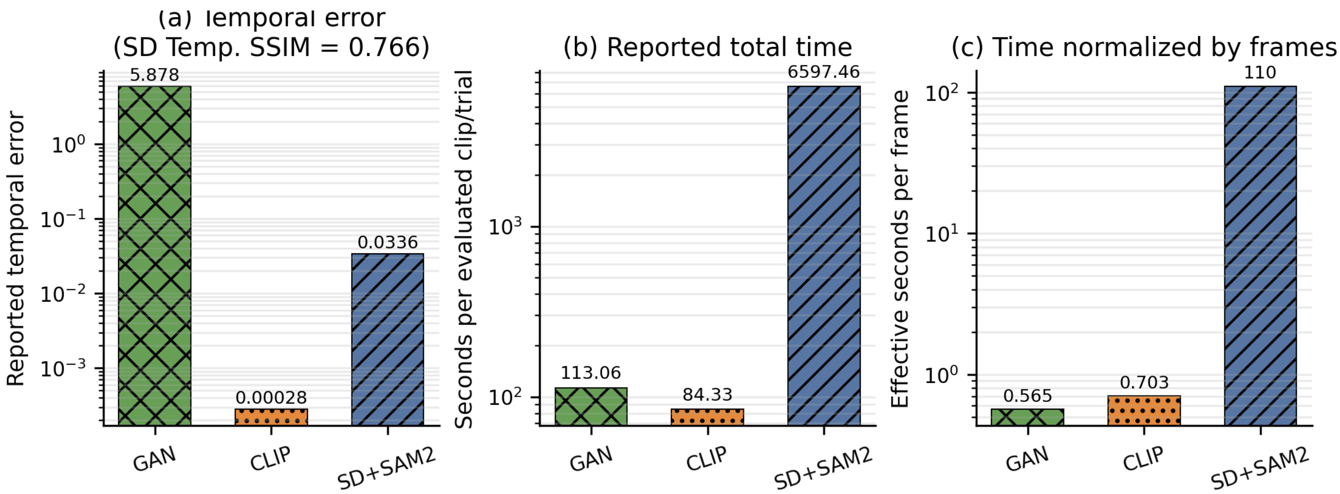

FIGURE 12. Temporal consistency and computational efficiency across the evaluated methods. The temporal-error values are complementary rather than directly identical: GAN and CLIP use background flow MAE, whereas Stable Diffusion + SAM2 uses frame-to-frame temporal L1. Likewise, GAN and CLIP report evaluation time, while Stable Diffusion + SAM2 reports generation time, so the timing panels indicate computational scale rather than a strictly controlled speed benchmark.

that never happened can spread rapidly online. This makes it easier to create and share false news, political propaganda, or harmful rumors. Even short, well-made inpainted videos can influence opinions before anyone checks if they are real.

C. PRIVACY VIOLATIONS IN SURVEILLANCE AND PERSONAL VIDEOS

Surveillance cameras often capture sensitive moments. Inpainting tools can remove people or objects from these recordings, making it possible to hide illegal activity or erase someone's presence without their knowledge. On the other hand, the same tools can add people who were never there. When applied to personal videos shared online, inpainting can expose private locations, reveal hidden details, or create embarrassing situations. Once a video is edited this way, it becomes very difficult to prove what was original and what was generated.

D. DATA OWNERSHIP AND TRAINING CONCERNS

Most generative video inpainting models are trained on large collections of public videos scraped from the internet. Many of these videos contain people who never gave permission for their faces or actions to be used in training data. This raises legal questions about who owns the right to use such footage. Similar to recent cases with large language models, creators of inpainting tools may face lawsuits if the training data includes copyrighted material or personal videos without consent. Users of these tools also need to know whether the output they create can be claimed as their own work or if it belongs partly to the original video owners.

E. MISUSE BY ORGANIZATIONS AND INDIVIDUALS

Companies and governments could use inpainting to "clean" surveillance videos before releasing them to the public, removing evidence of wrongdoing. Employees might use the tools to edit work-related videos and accidentally leak sensitive information. On the positive side, the same technology helps restore historical films or protect privacy by blurring faces automatically. The line between helpful editing and harmful manipulation is often thin and depends on the user's intent.

F. BIAS AND HALLUCINATIONS IN GENERATED CONTENT

Generative models sometimes produce biased results. If the training data contains mostly light-skinned people in certain settings, the inpainted areas may show unrealistic skin tones or clothing styles for other groups. Hallucinations—where the model invents details that do not match the real scene—can create flickering textures, wrong shadows, or impossible movements across frames. These errors are not just technical problems; they can make edited videos look suspicious or reinforce unfair stereotypes when used in media.

G. LEGAL AND REGULATORY CHALLENGES

Many countries are now creating laws against deepfakes and non-consensual video editing. However, enforcement is difficult because high-quality inpainting looks completely natural. Clear guidelines are needed on when inpainting is allowed (for example, in film restoration) and when it must be clearly labeled as edited content. Platforms that host videos may soon require automatic detection tools to mark inpainted clips, similar to how some sites label AI-generated images today. In summary, generative AI video inpainting brings exciting benefits for creativity and restoration, but it also creates serious risks to truth, privacy, and fairness. Developers, researchers, and policymakers must work together to build safeguards such as watermarking generated content, improving bias detection, and creating ethical usage rules. Only then can the technology be used responsibly while protecting individuals and society.

VII. OPEN CHALLENGES AND FUTURE DIRECTIONS

A. FROM VIDEO INPAINTING TO CONTROLLABLE VIDEO OBJECT REPLACEMENT

A natural next step beyond "filling holes" is *controllable* video editing: users do not only want a plausible completion, but they want to *change what appears in the video* (e.g., replace a logo on a shirt, change a billboard image, remove a watermark and insert new text, or swap an object with a specific reference). This setting adds new requirements on top of realism and temporal coherence:

- **Precise control:** the inserted content should match the user-provided target (image, logo, product, or text prompt) rather than an unconstrained "best guess."
- **Geometric consistency:** the replacement must respect perspective, scale, and scene motion across frames.
- **Photometric consistency:** lighting, shadows, and color tone should blend with the surrounding scene.
- **Long-range temporal stability:** the object identity should not drift over time, especially in long clips.

B. CONCRETE DIRECTION: CHANGING AN IMAGE IN A VIDEO WITH GENERATIVE MODELS

One concrete and practical future direction is to design a pipeline that takes a video and a user-provided *replacement image* and produces an edited video where that image is consistently integrated across all frames.

1) A practical, implementable pipeline

A strong baseline can be built by combining robust tracking with conditional generative inpainting:

1) **Region selection and stable masks.** Obtain a target mask M_t either from manual annotation on a keyframe with propagation, or from a segmentation/tracking model. Apply temporal smoothing to reduce jitter (e.g., mask warping with flow followed by morphological smoothing) because unstable masks directly produce flicker.
2) **Motion estimation and alignment.** Estimate motion between frames (optical flow or a planar homography for roughly planar surfaces such as screens and billboards). Use this motion to warp both the mask and the replacement image forward in time, producing a per-frame aligned target R_t.
3) **Conditioned generation.** Use a generative inpainting model to synthesize the masked region while conditioning on the aligned replacement R_t and optional structure guidance (edges, depth, surface normals, or semantic maps). In practice, diffusion inpainting with structural conditioning is attractive because it provides high-quality texture synthesis and can be guided to follow the replacement content.
4) **Temporal controls to prevent drift.** Enforce consistency by initializing each frame from a warped latent (or warped RGB) of the previous frame, reusing noise seeds within a temporal window, and/or using cross-frame attention. A complementary approach is to add an explicit warp-consistency loss during fine-tuning on video clips.
5) **Compositing and photometric blending.** After generation, blend the edited region into the original frame using feathered alpha masks and apply lightweight color matching to reduce seams. For difficult scenes, estimate illumination changes and apply them to the inserted image.
6) Generative Shape Replacement via Geometric Primitives
   Another promising direction is to combine generative AI with explicit geometric shape-abstraction techniques for object replacement. Instead of relying solely on a generative model to infer the geometry of an object, the target object can first be reconstructed using

compact parametric or primitive-based representations. Such representations provide an explicit description of the object's geometry that can subsequently be used to guide the generation of a replacement object.

One possible approach is to employ *Marching Primitives* [37], which approximates complex shapes represented by signed distance functions (SDFs) using a collection of geometric primitives. This representation provides a compact and interpretable abstraction of the target geometry and can capture important properties such as the object's overall dimensions, orientation, and local geometric structure. Alternatively, probabilistic superquadric recovery [38] can be used to estimate parametric superquadric primitives from noisy or incomplete point-cloud observations. Superquadrics are particularly attractive in this context because a relatively small set of parameters can describe a wide range of object geometries while providing explicit control over their size, shape, and pose.

These reconstructed geometric representations could then be integrated with a generative model to synthesize the replacement object. Specifically, the recovered primitives or superquadric parameters could serve as geometric conditioning signals or constraints for a generative model, allowing the generated object to conform to the geometry of the object being replaced. At the same time, the generative model could modify other properties of the object, such as its detailed geometry, appearance, or semantic category. This separation between geometric reconstruction and generative synthesis could provide substantially stronger geometric control than unconstrained generation.

Such a framework may be particularly useful when the replacement object must satisfy physical or spatial constraints within the scene. For example, the generated object could be constrained to maintain the original object's bounding dimensions, pose, support or contact regions, and interaction geometry while adopting a different shape or appearance. Combining explicit geometric representations with generative models therefore offers a potential pathway toward controllable object replacement in which the generated object is both semantically appropriate and geometrically compatible with the surrounding environment.

2) Key technical challenges

Even with a clear pipeline, several hard problems remain:

- **Occlusions and disocclusions:** when the target region becomes partially occluded, the system must avoid "painting" the replacement over foreground objects, and it must restore it correctly when it reappears.
- **Non-planar and deformable targets:** replacing content on cloth or curved surfaces requires stronger geometric reasoning than a simple homography.
- **Identity lock:** diffusion models can slightly alter textures or text across frames; preventing these subtle changes is essential for tasks like logo or text replacement.
- **Lighting and cast shadows:** realistic insertion often requires generating or adapting shadows and specular highlights that are consistent with the scene.

C. STRONGER CONDITIONING: STRUCTURE, GEOMETRY, AND 3D AWARENESS

Future work should increase controllability by adding more reliable conditioning signals:

- **Structure-first editing:** estimate edges, depth, pose and use them as constraints so the generated region respects geometry.
- **3D-aware representations:** for large viewpoint changes, reconstruct coarse scene geometry (depth or a neural 3D representation) and render a consistent target across frames before running a generative refinement stage.
- **Reference-based appearance constraints:** enforce similarity to a reference image/logo using feature-space losses or explicit "identity" embeddings, while still allowing the generator to adapt to lighting.

D. DATASETS AND EVALUATION FOR VIDEO REPLACEMENT

Benchmarking controllable replacement requires evaluation beyond standard inpainting metrics. Promising directions include:

- **Paired capture protocols:** capture the same scene twice to obtain partial ground truth for the edited region.
- **Synthetic-to-real validation:** generate synthetic training pairs where ground truth is available, then validate on real clips to assess robustness.
- **Task-specific metrics:** in addition to background preservation and warp error, measure *target fidelity* (how closely the inserted content matches the desired image and text) and *temporal stability* of that fidelity over time.

E. EFFICIENCY AND DEPLOYMENT

Many of the most realistic approaches are computationally heavy. Practical deployment motivates:

- **Faster sampling and distillation** to reduce diffusion steps while preserving quality.
- **Streaming processing** so long videos can be edited in chunks without quality discontinuities at chunk boundaries.
- **Model compression and caching** to achieve interactive performance.

F. RESPONSIBLE USE AND PROVENANCE

Finally, as video inpainting and replacement become more convincing, responsible deployment becomes essential. Future systems should support provenance (e.g., detectable watermarks or metadata), provide clear user consent mechanisms for sensitive content (faces, private scenes), and include safeguards against deceptive use cases.

VIII. CONCLUSION

This paper reviewed the evolution of video inpainting from classical propagation-based techniques to modern deep generative approaches, emphasizing the central role of temporal coherence in producing visually stable results. We organized the literature into four recurring method families—3D CNN-based, propagation-based, flow-guided, and attention-based—and discussed how recent generative models improve realism while introducing new challenges related to consistency and controllability.

Beyond the survey, we presented an implementable evaluation protocol and conducted empirical benchmarking on a custom real-world dataset designed to stress temporal stability under fast motion, large moving objects, occlusions, and illumination changes. The experiments highlight a common trade-off: methods that excel at per-frame detail often require explicit temporal controls (motion-aligned initialization, occlusion handling, or cross-frame attention) to prevent flicker, while strong propagation constraints can improve stability but may blur fine textures.

Overall, the results suggest that future progress will come from hybrid systems that combine robust mask tracking, reliable motion estimation, and controllable generative synthesis. In particular, a concrete next direction is *generative video object replacement*: changing an image in a video using generative models while preserving geometry, lighting, and long-range temporal identity. Our dataset and benchmarking framework provide a foundation for evaluating these systems in realistic conditions and for measuring progress toward reliable, efficient, and trustworthy video editing.

The CLIP-guided variant obtains almost perfect full-frame and edit-region SSIM, near-zero MAE, and extremely small LPIPS values. However, qualitative inspection shows that these scores are often achieved by retaining the original object rather than performing the requested replacement. Consequently, the CLIP results should be interpreted as strong preservation scores, not as evidence of superior editing success. The GAN baseline is more computationally economical than Stable Diffusion + SAM2, but it has much lower preservation scores than CLIP and the largest reported temporal error, with visible instability in difficult sequences.

Stable Diffusion + SAM2 produces the strongest qualitative substitutions, but this capability comes with a substantial computational cost and with larger source-deviation errors inside the edited region. Its high edit-region MAE and lower edit SSIM are expected when the original object is deliberately replaced by a semantically different target; therefore, these metrics should not be read as failures in isolation. In the category-level results of Table 7, the single human trial gives the strongest reported full-frame SSIM, background PSNR, edit-region SSIM, temporal L1, and temporal SSIM. Among the categories with three trials, cats show better temporal consistency than cars. Nevertheless, the human result is based on only one trial, and the category split in Table 7 differs from the newer seven-sequence re-aggregation, so these class-level trends remain descriptive rather than statistically conclusive.

Overall, no single preservation metric adequately captures the objective of video object replacement. A reliable evaluation should report background preservation, temporal consistency, computational cost, and semantic substitution success as separate dimensions. Future work should therefore add a direct semantic-success measure, such as prompt–region alignment or object-recognition accuracy within the edited mask, and should use a larger, balanced set of videos per category. This would distinguish methods that merely reproduce the input from methods that perform the intended replacement while preserving the surrounding scene and maintaining temporal coherence.

REFERENCES

[1] A. Ghanbari Talouki, M. Majdi, and S. A. Edalatpanah, "An introduction to various algorithms for video completion and their features: A survey," *Journal of Computer Sciences and Applications*, vol. 5, no. 1, pp. 1–10, 2017. [Online]. Available: https://doi.org/10.12691/jcsa-5-1-1

[2] H. Xiang, Q. Zou, M. A. Nawaz, X. Huang, F. Zhang, and H. Yu, "Deep learning for image inpainting: A survey," *Pattern Recognition*, vol. 134, p. 109046, 2023, article 109046. [Online]. Available: https://doi.org/10.1016/j.patcog.2022.109046

[3] X. Zhang, D. Zhai, T. Li, Y. Zhou, and Y. Lin, "Image inpainting based on deep learning: A review," *Information Fusion*, vol. 90, pp. 74–94, 2023. [Online]. Available: https://doi.org/10.1016/j.inffus.2022.08.033

[4] W. Quan, J. Chen, Y. Liu, D.-M. Yan, and P. Wonka, "Deep learning-based image and video inpainting: A survey," *International Journal of Computer Vision*, vol. 132, no. 7, pp. 2367–2400, 2024. [Online]. Available: https://doi.org/10.1007/s11263-023-01977-6

[5] O. Elharrouss, R. Damseh, A. N. Belkacem, E. Badidi, and A. Lakas, "Transformer-based image and video inpainting: current challenges and future directions," *Artificial Intelligence Review*, vol. 58, no. 4, p. 124, 2025, article number 124. [Online]. Available: https://doi.org/10.1007/s10462-024-11075-9

[6] S. Aghababyan, "Ai-powered video editing: Motion-aware object replacement using generative models," 2025, bachelor's thesis.

[7] W. Quan, J. Chen, Y. Liu, D. Yan, and P. Wonka, "Deep learning-based image and video inpainting: a survey," *International Journal of Computer Vision*, Jan 2024.

[8] H. Xiang, Q. Zou, M. A. Nawaz, X. Huang, F. Zhang, and H. Yu, "Deep learning for image inpainting: a survey," *Pattern Recognition*, vol. 134, p. 109046, Feb 2023.

[9] O. Elharrouss, R. Damseh, A. N. Belkacem, E. Badidi, and A. Lakas, "Transformer-based image and video inpainting: Current challenges and future directions," *Artificial Intelligence Review*, vol. 58, no. 4, Feb 2025.

[10] M. Bertalmio, G. Sapiro, V. Caselles, and C. Ballester, "Image inpainting," in *Proceedings of the 27th Annual Conference on Computer Graphics and Interactive Techniques*, ser. SIGGRAPH '00. USA: ACM Press/Addison-Wesley Publishing Co., 2000, p. 417–424.

[11] Y. Wang, Y.-C. Chen, X. Tao, and J. Jia, "Vcnet: A robust approach to blind image inpainting," in *Computer Vision – ECCV 2020*, A. Vedaldi, H. Bischof, T. Brox, and J.-M. Frahm, Eds. Cham: Springer International Publishing, 2020, pp. 752–768.

[12] S. S. Phutke, A. Kulkarni, S. K. Vipparthi, and S. Murala, "Blind image inpainting via omni-dimensional gated attention and wavelet queries," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, June 2023, pp. 1251–1260.

[13] H. Zhao, Y. Wang, Z. Gu, B. Zheng, and H. Zheng, "Context-aware mutual learning for blind image inpainting and beyond," *Expert Systems with Applications*, vol. 268, p. 126224, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0957417424030914

[14] J.-F. Cai, H. Ji, F. Shang, and Z. Shen, "Inpainting for compressed images," *Applied and Computational Harmonic Analysis*, vol. 29, no. 3, p. 368–381, Nov 2010.

[15] C. Dong, Y. Deng, C. C. Loy, and X. Tang, "Compression artifacts reduction by a deep convolutional network," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, December 2015.

[16] M.-J. Kwon, S.-H. Nam, I.-J. Yu, H.-K. Lee, and C. Kim, "Learning jpeg compression artifacts for image manipulation detection and localization," *International Journal of Computer Vision*, May 2022.

[17] Y. Gu, Y. Meng, X. Sun, J. Ji, W. Ruan, and R. Ji, "Mixed degradation image restoration via local dynamic optimization and conditional embedding," *rXiv preprint arXiv:2411.16217v1*, Nov 2024.

[18] W. Luo, H. Qin, Z. Chen, L. Wang, D. Zheng, Y. Li, Y. Liu, B. Li, and W. Hu, "Visual-instructed degradation diffusion for all-in-one image restoration," in *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, June 2025, pp. 12 764–12 777.

[19] D. Pathak, P. Krahenbuhl, J. Donahue, T. Darrell, and A. A. Efros, "Context encoders: Feature learning by inpainting," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016.

[20] C. Yang, X. Lu, Z. Lin, E. Shechtman, O. Wang, and H. Li, "High-resolution image inpainting using multi-scale neural patch synthesis," in *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4076–4084.

[21] Y. Zeng, Z. Lin, J. Yang, J. Zhang, E. Shechtman, and H. Lu, "High-resolution image inpainting with iterative confidence feedback and guided upsampling." Berlin, Heidelberg: Springer-Verlag, 2020, p. 1–17.

[22] C. Zheng, T.-J. Cham, and J. Cai, "Pluralistic image completion," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2019.

[23] C. Wang, Y. Zhu, and C. Yuan, "Diverse image inpainting with normalizing flow." Berlin, Heidelberg: Springer-Verlag, 2022, p. 53–69.

[24] D. Kim, S. Woo, J.-Y. Lee, and I. S. Kweon, "Deep video inpainting," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2019.

[25] Y. Zeng, J. Fu, and H. Chao, "Learning joint spatial-temporal transformations for video inpainting," in *Computer Vision – ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XVI*. Berlin, Heidelberg: Springer-Verlag, 2020, p. 528–543.

[26] Z. Wu, K. Li, H. Fan, and Y. Yang, "Drafting and revision: advancing high-fidelity video inpainting," in *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, ser. IJCAI '25, 2025.

[27] Y. Song, C. Yang, Z. Lin, X. Liu, Q. Huang, H. Li, and C.-C. Jay Kuo, "Contextual-based image inpainting: Infer, match, and translate," in *The European Conference on Computer Vision (ECCV)*, September 2018.

[28] S. Iizuka, E. Simo-Serra, and H. Ishikawa, "Globally and locally consistent image completion," vol. 36, no. 4, Jul. 2017.

[29] J. Yu, Z. Lin, J. Yang, X. Shen, X. Lu, and T. S. Huang, "Generative image inpainting with contextual attention," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2018.

[30] G. Liu, F. A. Reda, K. J. Shih, T.-C. Wang, A. Tao, and B. Catanzaro, "Image inpainting for irregular holes using partial convolutions," in *Proceedings of the European Conference on Computer Vision (ECCV)*, September 2018.

[31] Y.-G. Shin, M.-C. Sagong, Y.-J. Yeo, S.-W. Kim, and S.-J. Ko, "Pepsi++: Fast and lightweight network for image inpainting," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 1, p. 252–265, Jan 2021.

[32] Y. Wexler, E. Shechtman, and M. Irani, "Space-time completion of video," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 29, no. 3, p. 463–476, Mar 2007.

[33] M. Granados, K. I. Kim, J. Tompkin, J. Kautz, and C. Theobalt, "Background inpainting for videos with dynamic objects and a free-moving camera," in *Computer Vision – ECCV 2012*, A. Fitzgibbon, S. Lazebnik, P. Perona, Y. Sato, and C. Schmid, Eds., vol. 7572. Berlin, Heidelberg: Springer, 2012, pp. 682–695.

[34] A. Newson, A. Almansa, M. Fradet, Y. Gousseau, and P. Pérez, "Video inpainting of complex scenes," *SIAM Journal on Imaging Sciences*, vol. 7, no. 4, p. 1993–2019, Jan 2014.

[35] J. Huang, S. B. Kang, N. Ahuja, and J. Kopf, "Temporally coherent completion of dynamic video," *ACM Transactions on Graphics*, vol. 35, no. 6, p. 1–11, Nov 2016.

[36] R. Komatsu and T. Gonsalves, "Comparing u-net based models for denoising color images," *AI*, vol. 1, no. 4, pp. 465–486, 2020. [Online]. Available: https://www.mdpi.com/2673-2688/1/4/29

[37] W. Liu, Y. Wu, S. Ruan, and G. S. Chirikjian, " Marching-Primitives: Shape Abstraction from Signed Distance Function ," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Los Alamitos, CA, USA: IEEE Computer Society, Jun. 2023, pp. 8771–8780. [Online]. Available: https://doi.ieeecomputersociety.org/10.1109/CVPR52729.2023.00847

[38] ——, "Robust and accurate superquadric recovery: a probabilistic approach," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 2666–2675.

STEPAN GOYUNYAN is an AI/ML Engineering Intern at an applied AI research lab based in Yerevan, Armenia. He is a graduate of PhysMath School in Yerevan (shortlisted as one of ten finalists worldwide for the 2023 World's Best School Prizes). His technical foundation was built through competitive mathematics and physics olympiads, self-directed machine learning study, and training programs including the Armenian Code Academy (ACA) and the IOAI Preparatory Club.

His research interests include natural language processing, representation learning, retrieval systems, and physical AI (vision-language-action models). He will be starting his Bachelor's degree in Computer Science at the American University of Armenia (AUA) this year.

ORCID: 0009-0003-1292-1237.

NAREK MELIKSETYAN is a Forward Deployed Engineer and team lead at Starshunt AI, where he builds and delivers production AI systems and leads his own team.

He is a graduate of PhysMath School in Yerevan (shortlisted as one of ten finalists worldwide for the 2023 World's Best School Prizes). His technical foundation was built through competitive mathematics and physics olympiads, a full-scholarship place on the Armenian Code Academy (ACA) machine learning program, and continuing study with the IOAI Preparatory Program. He previously served as a Research Assistant in the Graphene Materials Group at the A. Alikhanyan National Science Laboratory, synthesizing graphene by liquid-phase exfoliation and running FTIR and Raman spectroscopy for an ongoing proton-irradiation study that feeds into a peer-reviewed publication.

His research interests include computer vision, video understanding, and the reliability and evaluation of large language model systems. He co-founded the PhysMath Academic Initiative, a student-led mentoring program whose five mentors support more than ninety students in mathematics, physics, and computer science, with tracked outcomes including sixteen prevented dropouts and more than ten admissions to leading universities.

ORCID: 0009-0009-6863-1312.

ARTUR TSATURYAN is an independent AI researcher and student at the Evrika STEM High School in Vanadzor, Armenia, participating in the Generation AI program. His work focuses on machine learning, computer vision architectures, and large language model optimization. Previously, he served as an AI Engineer at OMNICODE AI, contributing to the design and deployment of neural network systems. His recent development projects include computer vision solutions for wearable assistive technology and synthetic data modeling for clinical applications.

ORCID: 0009-0007-0788-2287.

ELIZAVETA LABAZANOVA is an AI Training Data Specialist contributing to a Sber project focused on developing artificial intelligence models capable of generating realistic human video content. She graduated from Gymnasium No. 14, Grozny, Russia, where she developed a foundation in Python programming and won first place in the "KIT" information technology competition. She has also participated in international volunteer programs and delivered presentations at events in

Kabardino-Balkaria commemorating the 75th anniversary of the victory in World War II. Her professional interests lie at the intersection of business and artificial intelligence, a field she plans to pursue through a bachelor's degree.

ORCID: 0009-0004-2765-0000.

ZANGIR IKLASSOV has a career spanning academic research, teaching, and industry innovation. Before assuming his current faculty position, he served as a postdoctoral researcher at Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), where he contributed to research projects in optimization and large language model reasoning. He also led research efforts for ADNOC's EnergyAI project, developing domain-specific artificial intelligence solutions. His industry experience includes co-founding Smart System Technologies Ltd., a startup specializing in road damage detection, and holding engineering roles at ReLive Intelligence Ltd. and Computer Vision Technologies Ltd., where he developed AI algorithms for large-scale computer vision applications. In academia, he has supported courses in Advanced Machine Learning and Deep Learning, mentored students, and delivered invited talks, including a research seminar at the University of Oxford.

ORCID: 0000-0002-2835-990X.

JORGE ALEJANDRO AMADOR HERRERA earned his Ph.D. in Applied Mathematics and Computational Sciences from King Abdullah University of Science and Technology (KAUST), where he received the KAUST Fellowship and the CEMSE Dean's List Award for academic excellence. His doctoral research focused on physically based simulations of weather phenomena, combining computer graphics, scientific computing, and machine learning. He also held research positions at Adobe and CERN. At Adobe, he developed real-time frameworks for digital humans, including techniques for hair simulation and deformable distance fields. At CERN, he contributed to data-driven anomaly detection methods for high-energy physics experiments conducted using the Compact Muon Solenoid (CMS) detector. Earlier in his career, he conducted research in cosmology, molecular dynamics, and scheduling optimization at the National Autonomous University of Mexico (UNAM).

ORCID: 0000-0002-4607-4772.